

# On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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## Abstract

We study  $G_n = \gcd(d_n a_n, d_n b_n)$  for the Apéry sequences  $(a_n), (b_n)$  with  $d_n = \text{lcm}(1, \dots, n)^3$ . We prove *unconditionally* that  $G_n = e^{o(n)}$  for a set of positive integers of natural density 1, with the quantitative bound  $\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2)$  for every  $\varepsilon > 0$ ; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. The key ingredient is a new bound  $Z(p) = O(p^{2/3})$  on the zero-count function  $Z(p) = \#\{j < p : p \mid b_j\}$ , derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes  $5 \leq p \leq 10^6$  gives  $Z(p) \leq 12$  with mean  $\approx 1$ . The pair count  $Z(p)/2$  fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ , so  $Z(p)$  is almost certainly unbounded. Under the weaker Cesàro hypothesis  $\sum_{p \leq x} Z(p) \ll \pi(x)$  (Hypothesis  $\bar{Z}$ ), which the Poisson model supports, we prove  $\log G_n = O(\sqrt{n})$  for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to  $O(N^{1/3})$  and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from  $O(N^{3/2})$  to  $O(N^{10/7}/\log N)$  unconditionally, and to  $O(N^{4/3}/\log N)$  under Hypothesis  $\bar{Z}$ .

## 1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5) u_n - n^3 u_{n-1}, \quad n \geq 1, \quad (1)$$

with  $(a_0, a_1) = (0, 6)$  and  $(b_0, b_1) = (1, 5)$ . Here  $b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$  is the  $n$ -th Apéry number. Set  $d_n = \text{lcm}(1, \dots, n)^3$ ; Apéry [3] showed  $d_n a_n, d_n b_n \in \mathbb{Z}$ . Define  $G_n = \gcd(d_n a_n, d_n b_n)$ . Write  $P(n) = 34n^3 + 51n^2 + 27n + 5$  for the middle coefficient.

## 2 The Wronskian identity

**Lemma 1.** *For all  $n \geq 1$ ,  $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ .*

*Proof.* The Casorati determinant satisfies  $W_{n+1}/W_n = n^3/(n+1)^3$  from (1). Since  $W_1 = a_1 b_0 - a_0 b_1 = 6$ , telescoping gives  $W_n = 6/n^3$ .  $\square$

**Corollary 2.** *For every prime  $p \geq 5$ ,*

$$v_p(G_n) \leq 3 \lfloor \log_p n \rfloor. \quad (2)$$

*Proof.*  $G_n$  divides  $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$ . For  $p \geq 5$ ,  $v_p(6) = 0$  and  $v_p(d_n/n^3) = 3 \lfloor \log_p n \rfloor - 3v_p(n) \leq 3 \lfloor \log_p n \rfloor$ .  $\square$

### 3 Small primes: unconditional bound

**Proposition 3.**  $\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n}).$

*Proof.* By (2),  $v_p(G_n) \log p \leq 3 \log n$  for every prime  $p \geq 5$ . Primes  $p = 2, 3$  contribute at most  $O(\log n)$  each. There are  $O(\sqrt{n}/\log n)$  primes up to  $\sqrt{n}$ , so the sum is at most  $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$ .  $\square$

### 4 The multiplicative Lucas congruence

**Theorem 4** (Gessel [13]). *For every prime  $p$  and  $n = \sum_{i=0}^s n_i p^i$  in base  $p$ ,*

$$b_n \equiv \prod_{i=0}^s b_{n_i} \pmod{p}. \quad (3)$$

This follows from the combinatorial identity  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$  and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [17] and McIntosh [16].

### 5 The zero-count function

**Definition 5.** For a prime  $p \geq 5$ , define  $Z(p) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv 0 \pmod{p}\}$ .

#### 5.1 Unconditional bound

**Lemma 6** (No consecutive zeros). *For every prime  $p \geq 5$  and every  $j \in \{0, \dots, p-2\}$ ,  $b_j$  and  $b_{j+1}$  cannot both vanish modulo  $p$ .*

*Proof.* Suppose  $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$  with  $0 \leq m \leq p-2$ . The recurrence (1) at  $n = m$  gives  $(m+1)^3 b_{m+1} = P(m) b_m - m^3 b_{m-1}$ , so  $m^3 b_{m-1} \equiv 0 \pmod{p}$ . Since  $1 \leq m \leq p-2$  implies  $p \nmid m$ , we get  $b_{m-1} \equiv 0$ . Iterating gives  $b_0 \equiv 0 \pmod{p}$ , contradicting  $b_0 = 1$ .  $\square$

**Lemma 7** (Gap polynomial). *Fix  $h \geq 2$ . Suppose  $b_m \equiv 0 \pmod{p}$  and  $m+h \leq p-1$ . Then*

$$b_{m+h} \equiv C_h(m) b_{m+1} \pmod{p}, \quad (4)$$

where  $C_h(m)$  is a rational function of  $m$  whose numerator (after clearing the denominator  $\prod_{j=1}^{h-1} (m+j+1)^3$ ) is a polynomial of degree exactly  $3(h-1)$  with positive leading coefficient.

In particular, if  $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$ , then by Lemma 6  $b_{m+1} \not\equiv 0$ , so  $C_h(m) \equiv 0 \pmod{p}$ , and  $m$  is constrained to at most  $3(h-1)$  residue classes modulo  $p$ , provided  $C_h$  is not identically zero over  $\mathbb{F}_p$ .

*Proof.* We iterate the recurrence from the initial condition  $(b_m, b_{m+1}) = (0, b_{m+1})$ . At step  $n = m+1$ :  $b_{m+2} = P(m+1) b_{m+1} / (m+2)^3$ , since  $b_m = 0$  kills the second term. At step  $n = m+j$  for  $j \geq 2$ :  $b_{m+j+1}$  is a  $\mathbb{Q}(m)$ -linear combination of  $b_{m+j}$  and  $b_{m+j-1}$ , both already rational multiples of  $b_{m+1}$ . The leading coefficient of  $P(n)$  is 34 and the subtracted term has leading coefficient 1, so at each step the degree in  $m$  increases by 3 and the leading coefficient remains positive. After  $h-1$  steps the numerator has degree  $3(h-1)$ .

For  $h = 2$ :  $C_2(m) = P(m+1)/(m+2)^3$ , with numerator  $P(m+1) = (2m+3)(17m^2+51m+39)$ , a cubic with leading coefficient 34.  $\square$

**Lemma 8** (Nonvanishing over  $\mathbb{F}_p$ ). *For every prime  $p \geq 7$  and every  $2 \leq h \leq p$ , the numerator of  $C_h$  is not the zero polynomial modulo  $p$ .*

*Proof.* Write  $N_h$  for the numerator of  $C_h$  (a polynomial of degree  $3(h-1)$ ), satisfying  $N_{h+1}(m) = P(m+h)N_h(m) - (m+h)^6 N_{h-1}(m)$ , with  $N_1 = 1$  and  $N_2 = P(m+1)$ .

*First evaluation.* Equation (4) at  $m = -1$  (where  $b_{-1} = 0$ ,  $b_0 = 1$ ) gives  $C_h(-1) = b_{h-1}$ , hence  $N_h(-1) = b_{h-1} \cdot ((h-1)!)^3$ .

*Second evaluation.* Setting  $m = -2$  in the recurrence gives  $N_{h+1}(-2) = P(h-2)N_h(-2) - (h-2)^6 N_{h-1}(-2)$ , with  $N_1(-2) = 1$  and  $N_2(-2) = P(-1) = -5$ . By induction using the Apéry recurrence at  $n = h-2$ :

$$N_h(-2) = -5b_{h-2}((h-2)!)^3 \quad (h \geq 2). \quad (5)$$

*Conclusion.* If  $N_h \equiv 0 \pmod{p}$  as a polynomial ( $2 \leq h \leq p$ ,  $p \geq 7$ ), then  $N_h(-1) \equiv N_h(-2) \equiv 0$ , forcing  $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod{p}$  (since  $p \nmid 5$  and  $p \nmid (h-1)!, (h-2)!$ ). These are consecutive zeros, contradicting Lemma 6.  $\square$

**Proposition 9.** *For every prime  $p \geq 7$ ,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \frac{3}{2}(H-1)(H-2) + 1, \quad (6)$$

where  $H$  is any integer with  $2 \leq H \leq p$ . Optimizing  $H = \lfloor (p/3)^{1/3} \rfloor + 1$  gives  $Z(p) \leq \frac{3^{4/3}}{2} p^{2/3} + O(p^{1/3}) = O(p^{2/3})$  unconditionally for all primes. In particular,  $Z(p) = o(p)$ .

*Proof.* Partition  $\{0, \dots, p-1\}$  into  $\lceil p/H \rceil$  consecutive blocks of length at most  $H$ . In each block, mark the first zero; all other zeros have gap  $h \in \{2, \dots, H-1\}$  to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap- $h$  pair constrains  $m$  to at most  $3(h-1)$  residue classes mod  $p$  (the denominator factors  $(m+j+1)^3$  are nonzero since  $0 \leq m+j+1 \leq p-1$ ). By Lemma 8,  $C_h \not\equiv 0$  over  $\mathbb{F}_p$ , so the numerator roots are at most  $3(h-1)$ . Summing:

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \frac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real  $H > 0$  occurs at  $H = (p/3)^{1/3}$ , giving the bound. For  $p = 2, 3, 5$ :  $Z(p) = 0$  (since  $b_j$  for  $j < 5$  is coprime to  $2, 3, 5$ ).  $\square$

**Corollary 10** (Sub-interval zero count). *For every prime  $p \geq 7$  and every interval  $I \subseteq \{0, \dots, p-1\}$ ,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

*Proof.* Apply the proof of Proposition 9 to the sub-interval  $I$  in place of  $\{0, \dots, p-1\}$ : partition  $I$  into  $\lceil |I|/H \rceil$  blocks of length  $H$ , bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize.  $\square$

*Remark 11.* The leading coefficient of the numerator of  $C_h$  is  $U_{h-1}(17)$ , the Chebyshev polynomial of the second kind evaluated at 17, satisfying  $\ell_1 = 1$ ,  $\ell_{h+1} = 34\ell_h - \ell_{h-1}$ . Let  $N_h(m)$  denote the cleared numerator of  $C_h$ . The identity  $P(-u-1) = -P(u)$  (evident from  $P(n) = (2n+1)(17n^2 + 17n + 5)$ ) propagates through the recurrence to give  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ , verified computationally for  $h \leq 200$ . For even  $h$ ,  $N_h$  is an *odd* polynomial in  $(m + (h+1)/2)$ , forcing the structural factor  $(2m+h+1)$ . At  $h = 2$ : factor  $(2m+3)$  in  $P(m+1) = (2m+3)(17m^2 + 51m + 39)$ . The computational data (Section 10) suggests  $Z(p) = O(1)$ , far stronger than  $O(p^{2/3})$ .

*Remark 12* (Content restriction). A stronger nonvanishing holds: if  $p \geq 7$  divides the content of  $N_h$  (the gcd of all its coefficients), then  $h \geq 2p$ . Indeed,  $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} ((r-1)!)^3 ((h-r)!)^3$  for  $1 \leq r \leq h$  (by induction from the Apéry recurrence); choosing  $r = 1, 2$  and using  $p \nmid (h-1)!, (h-2)!, 5$  for  $h \leq p$  recovers Lemma 8, while the range  $p < h < 2p$  is excluded similarly via  $r = h - p + 1$ . Setting  $\beta_n = (n!)^3 b_n$  and  $F(z) = \sum_{n \geq 0} b_n z^n$ , the endpoint factorization packages into the double generating function  $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u) F(v)$ , showing that the boundary array is a rank-one tensor.

*Remark 13* (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$|\text{Res}(N_h, N_{h+1})| = \prod_{j=1}^{h-1} (j!^3 b_j)^6. \quad (7)$$

This follows from the recursion  $\text{Res}(N_h, N_{h+1}) = N_h(-h)^6 \text{Res}(N_{h-1}, N_h)$  (since  $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$ ), combined with  $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1} ((h-1)!)^3 b_{h-1}$  (the reflection identity and the first evaluation from Lemma 8). Verified for  $h = 2, 3, 4, 5$ ; the cases  $h = 2, 3, 4$  give  $\text{Res}(N_2, N_3) = -5^6$ ,  $\text{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$ ,  $\text{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$ .

A key consequence:  $p \mid \text{Res}(N_h, N_{h+1})$  if and only if  $p \mid b_j$  for some  $1 \leq j \leq h-1$  (when  $p > h$ , so factorials are coprime to  $p$ ). In particular,  $N_2$  and  $N_3$  have disjoint root sets modulo  $p$  for all  $p \geq 7$  (since  $5^6 \neq 0$ ).

*Remark 14* (Sharpness of the bound). The exponent  $2/3$  in Proposition 9 is optimal for any argument using only the degree bounds  $\deg N_h = 3(h-1)$ . The exact extremal problem  $\max\{\sum r_h : r_h \leq 3(h-1), \sum h r_h \leq p\}$  has value  $\frac{3}{2} p^{2/3} + O(p^{1/3})$ , with leading constant  $3/2$  achieved by filling gap sizes  $h = 2, 3, \dots, H$  to capacity and using the remaining space at  $h = H+1$ , where  $H \sim p^{1/3}$ . Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield  $O(p^{1/2})$ : an abstract polynomial model on  $\sim p^{2/3}$  points can satisfy  $\deg F_h \leq 3(h-1)$  for all relevant  $h$ . Improving the exponent requires genuinely arithmetic input about the family  $(N_h)$ .

*Remark 15* (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll’s condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m) N_h(m+1) - N_{h+1}(m) N_{h+1}(m+1) = - \prod_{j=2}^{h+1} (m+j)^6. \quad (8)$$

This is an identity in  $\mathbb{Z}[m]$ , verified for  $h \leq 7$ . A consequence: for  $m \not\equiv -j \pmod{p}$  ( $2 \leq j \leq h+1$ ), the pairs  $(m, h+2)$  and  $(m+1, h)$  cannot both lie in the zero relation  $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$ , since  $N_{h+1}(m) N_{h+1}(m+1)$  would equal a nonzero product of sixth powers. After a meromorphic gauge  $\tau_h(x) = N_h(x)/d_h(x)$  satisfying  $d_{h+2}(x) d_h(x+1) = \prod_{j=2}^{h+1} (x+j)^6$ , this becomes an exact  $SL_2$ -tiling:  $\tau_{h+1}(x) \tau_{h+1}(x+1) - \tau_{h+2}(x) \tau_h(x+1) = 1$ , equivalently an  $A_1$   $T$ -system (discrete Liouville equation). A zero of  $\tau$  forces its transverse neighbors to be units (*singularity confinement*), but abstract  $SL_2$ -tilings can have zeros of density  $1/2$  (e.g.,  $q_n = 0, 1, 0, -1, \dots$  with  $q_n^2 - q_{n-1} q_{n+1} = 1$ ), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

*Remark 16* (Projective orbit reformulation). Let  $B_n = n!^3 b_n$  and  $D_n$  be the second solution of the renormalized recurrence  $w_{n+1} = P(n) w_n - n^6 w_{n-1}$  with  $(D_0, D_1) = (0, 1)$ . Their Casoratian is  $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$ , nonzero for  $n < p$ . Define the *projective Apéry orbit*  $\pi : \{0, \dots, p-1\} \rightarrow \mathbb{P}^1(\mathbb{F}_p)$  by  $\pi_n = [B_n : D_n]$ . For  $0 \leq m < m+h < p$ ,  $N_h(m) \equiv 0 \pmod{p}$  if and only if  $\pi_m = \pi_{m+h}$ . Thus the zero-count  $Z(p)$  equals the number of distinct collisions in the orbit  $\pi$ . The reflection

symmetry  $P(-1-n) = -P(n)$  implies  $b_{p-1-n} \equiv b_n \pmod{p}$ , making the orbit a palindrome:  $\pi_n = \pi_{p-1-n}$ . In particular,  $Z(p)$  is even except at the non-ordinary primes  $p \mid a_p(f)$  (at most two primes  $p \leq 10^5$ ).

*Remark 17* (Galois structure). For each fixed  $h$ , the Galois group of the splitting field of  $N_h$  over  $\mathbb{Q}$  is not the full symmetric group: the reflection  $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$  forces roots into pairs  $\{r, -r-h-1\}$ , and  $\text{Gal}(N_h) \cong C_2 \wr S_m$  (hyperoctahedral group) with  $m = \lfloor 3(h-1)/2 \rfloor$  (confirmed exactly for  $h \leq 5$ ,  $2 \leq h \leq 9$  by obstruction tests). By the Chebotarev density theorem, the average number of  $\mathbb{F}_p$ -roots of  $N_h$  is 1 for odd  $h$  and 2 for even  $h$ . This prediction is verified: for 262 primes in  $[1000, 3000]$ , the measured averages are 0.98, 2.03, 1.11, 2.07, 1.11, 1.88 for  $h = 3, \dots, 8$ .

**Proposition 18** (Hyperoctahedral Poisson law). *Assume  $\text{Gal}(N_h) \cong C_2 \wr S_m$  with  $m = \lfloor 3(h-1)/2 \rfloor$ . Let  $Y_h(p)$  be the number of non-central reflection-pairs of roots of  $N_h$  that split completely over  $\mathbb{F}_p$ . Then for each fixed  $k \leq m$ , the factorial moments in the prime-aspect satisfy*

$$\lim_{x \rightarrow \infty} \frac{1}{\pi(x)} \sum_{p \leq x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where  $(Y)_k = Y(Y-1) \cdots (Y-k+1)$ . Since these match the factorial moments of  $\text{Poisson}(1/2)$  for all  $k$ , the prime-aspect distribution of  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$  as  $h \rightarrow \infty$ .

*Proof.* In  $G = C_2 \wr S_m = \{(\sigma, \varepsilon) : \sigma \in S_m, \varepsilon \in \{\pm 1\}^m\}$ , a positive fixed point of  $g = (\sigma, \varepsilon)$  is an index  $i$  with  $\sigma(i) = i$  and  $\varepsilon_i = +1$ . The Frobenius at  $p$  acts on the  $m$  reflection-pairs, and  $Y_h(p)$  equals the number of positive fixed points. For any  $k$  distinct indices  $i_1, \dots, i_k$ , the fraction of  $g \in G$  fixing all  $k$  positively is

$$\frac{2^{m-k} (m-k)!}{2^m m!} = \frac{1}{2^k (m)_k}.$$

Summing over all  $(m)_k$  ordered  $k$ -tuples,  $\mathbf{E}_{g \in G}[(Y(g))_k] = (1/2)^k$ , which matches the factorial moments of  $\text{Poisson}(1/2)$ . The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all  $k \leq m$ ,  $Y_h(p) \rightarrow \text{Poisson}(1/2)$  as  $m \rightarrow \infty$ .  $\square$

*Remark 19* (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed  $h$  and varying  $p$ , the root-pair count  $Y_h(p)$  converges to  $\text{Poisson}(1/2)$ . The zero-count  $Z(p)$  aggregates contributions from all gaps  $h = 1, \dots, p-1$  at a *fixed* prime  $p$ , so the orbit-aspect independence needed to deduce  $Z(p)/2 \sim \text{Poisson}(1/2)$  is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

*Remark 20* (GRH-effective error). For each fixed  $k$ , the  $k$ -th factorial moment does not require the full splitting field  $K_h$  (of degree  $2^m m!$ , double-exponential in  $h$ ). Let  $E_{h,k}$  be the fixed field of the stabilizer of an ordered signed  $k$ -tuple with distinct underlying reflection pairs; then  $[E_{h,k} : \mathbb{Q}] = 2^k (m)_k = O_k(h^k)$ . Under GRH for  $\zeta_{E_{h,k}}$ ,

$$\frac{1}{\pi(x)} \sum_{\substack{p \leq x \\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where  $D_{h,k} = |\text{Disc}(E_{h,k})|$ . The Apéry recurrence gives the coefficient-height bound  $\log \|N_h\|_1 = O(h \log h)$ , from which  $\log D_{h,k} \ll_k h^{k+2} \log h$ . Thus an  $O(h^{-C})$  error holds in the polynomial range  $x \geq h^{2k+4+2C+\varepsilon}$ , for each fixed  $k$ .

At  $x = 10^6$  with  $h = 10$  ( $m = 13$ ), the degree contributions alone for  $k = 1, 2, 3$  are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

## 5.2 Dual bounds: the restart identity and column estimate

The bound  $Z(p) = O(p^{2/3})$  controls zeros along a *fixed row* (varying  $m$  for a given gap  $h$ ). A dual bound controls zeros along a *fixed column* (varying  $h$  for a given starting point  $x_0$ ).

**Lemma 21** (Restart identity). *Suppose  $N_r(x_0) = 0$  and  $N_{r+1}(x_0) \neq 0$  for some  $x_0 \in \mathbb{F}_p$  and  $r \geq 1$ . Then for every  $d \geq 0$ ,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \quad (9)$$

*In particular, if  $N_{r+d}(x_0) = 0$  as well, then  $N_d(x_0 + r) = 0$ .*

*Proof.* The sequence  $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$  has  $Y_0 = 0$ ,  $Y_1 = 1$ , and satisfies  $Y_{d+1} = P((x_0+r)+d) Y_d - ((x_0+r)+d)^6 Y_{d-1}$ , which is the defining recurrence for  $N_d(x_0 + r)$ .  $\square$

**Proposition 22** (Column bound). *For every  $x_0 \in \mathbb{F}_p$  and every interval  $I$  of  $H$  consecutive  $h$ -values with  $H \leq p$ ,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \frac{3^{4/3}}{2} H^{2/3} + O(H^{1/3} + 1). \quad (10)$$

*Proof.* Split  $I$  at the unique transition  $h$  where  $x_0 + h \equiv 0 \pmod{p}$  (if it exists in  $I$ ), creating at most two nonsingular pieces plus an  $O(1)$  additive cost.

On a nonsingular piece, order the zero levels. For threshold  $L \geq 2$ : a gap of length  $d \leq L$  between consecutive zeros at levels  $r < r + d$  gives  $N_d(x_0 + r) = 0$  by Lemma 21. For fixed  $d$ , there are at most  $3(d-1)$  values of  $x_0 + r$  satisfying this (by Lemma 8 and  $\deg N_d = 3(d-1)$ ). Gaps longer than  $L$  contribute at most  $H/(L+1)$  zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^L 3(d-1) + O(1) = \frac{H}{L+1} + \frac{3}{2}L(L-1) + O(1).$$

Optimizing  $L \asymp (H/3)^{1/3}$  gives the stated bound.  $\square$

Computationally, the column count at mesoscopic scale  $H = \lfloor \sqrt{p} \rfloor$  is far below the  $O(H^{2/3})$  bound: for all primes  $p \leq 100,003$  and all  $x_0$ ,  $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$ . For  $p \geq 10,007$ , the maximum is  $\leq 1$ .

*Remark 23* (Rectangular incidence estimate). The row bound  $|R_h| \leq 3(h-1)$  and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p, 2 \leq h \leq H, N_h(x) \equiv 0\} \leq \min\{H^2, pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at  $H \sim p^{3/4}$ . Note that a uniform bound  $R_p(H) \leq CH$  for all primes is *false*: by the Chebotarev density theorem, for each fixed  $H$ , infinitely many primes  $p > H^2$  have every  $N_h$  ( $2 \leq h \leq H$ ) splitting completely over  $\mathbb{F}_p$ , giving  $R_p(H) = \frac{3}{2}H(H-1)$ . The correct target is the *mesoscopic* estimate  $R_p(\lfloor \sqrt{p} \rfloor) = O(\sqrt{p})$ , where the split-prime obstruction does not apply because  $H = \sqrt{p}$  grows with  $p$ . More generally, if  $R_p(H) \ll H^\beta$  then  $Z(p) \leq p/H + R_p(H)$  is optimized at  $H \sim p^{1/(\beta+1)}$ , giving  $Z(p) \ll p^{\beta/(\beta+1)}$ : the current  $\beta = 2$  gives  $2/3$  and  $\beta = 1$  yields  $1/2$ . Reflection forces  $R_p(H) \geq H/2$  from central roots, so  $\beta = 1$  is the intrinsic limit of this first-moment method.

### 5.3 The continuant Bezout identity

Write  $S_d(x) = \prod_{j=2}^d (x+j)$ .

**Proposition 24** (Bezout identity). *For integers  $2 \leq d < e$ ,*

$$N_{e-1}(x+1) N_d(x) - N_{d-1}(x+1) N_e(x) = S_d(x)^6 N_{e-d}(x+d). \quad (11)$$

*In particular, after saturating the cut-edge divisor  $S_d$ ,*

$$(N_d, N_e) : S_d^\infty = (N_d, N_{e-d}(x+d)) : S_d^\infty.$$

*Proof.* The continuant addition formula (splitting the interval  $[1, e-1]$  at position  $d$ ) gives  $N_e = N_d N_{e-d+1}(x+d-1) - (x+d)^6 N_{d-1} N_{e-d}(x+d)$ . The Dodgson identity at index  $d-2$  gives  $N_{d-1}(x+1) N_{d-1} - N_{d-2}(x+1) N_d = \prod_{j=2}^{d-1} (x+j)^6$ . Combining these two identities yields (11). Verified computationally for all  $2 \leq d < e \leq 9$ .  $\square$

**Corollary 25** (Separated-block criterion). *Away from the cut-edge divisor  $S_d(x) = 0$ ,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod{p} \iff N_d(\alpha) \equiv N_{e-d}(\alpha+d) \equiv 0 \pmod{p}.$$

*In particular,  $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \leq 3(\min(d, e-d) - 1)$ , which improves the naive  $3(d-1)$  when  $e-d < d$ .*

**Remark 26** (Separated-block resultant). Define  $\mathcal{R}_{d,r} = \text{Res}_x(N_d(x), N_r(x+d))$ . The Bezout identity gives

$$|\text{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} ((j!)^3 b_j)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ( $e = d+1$ ),  $N_1 = 1$  gives  $\mathcal{R}_{d,1} = 1$ , recovering the adjacent resultant formula. For nonadjacent indices,  $\mathcal{R}_{d,r}$  is genuinely new. The reflection identity gives  $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$ . The subtraction step does *not iterate*: after one application,  $N_d(x)$  and  $N_{e-d}(x+d)$  live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For  $d = 2$ , writing  $P(y) = (2y+1)Q(y)$  with  $Q(y) = 17y^2 + 17y + 5$  and  $\theta$  a root of  $Q$ , the norm decomposition  $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)} N_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$  (where  $G_r = N_r(x+1)$ ) gives  $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$ . At  $p = 71$ :  $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$  with  $-5/2 \notin \{-1, -2, -3, -4\}$ , a nonboundary common root inside the mesoscopic range ( $4^3 < 71$ ). This shows the boundary-only gcd statement is *false*.

**Remark 27** (Collision energy and the  $H^{1/2}$  barrier). The column exponent  $2/3$  is the natural barrier for the consecutive-gap argument: even if each gap of length  $d$  has only  $d$  (rather than  $3(d-1)$ ) admissible starts, the sum  $\sum_{d \leq L} d \asymp L^2$  still gives  $H/L + L^2$ , optimized at  $H^{2/3}$ . To achieve  $H^{1/2}$ , one needs the aggregate collision energy  $\mathcal{E}_p(H) = \sum_{d+r \leq H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$ , which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The  $A_1$  T-system  $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$  admits tame solutions with zero density  $1/2$ , e.g.  $\tau_{2k} = 0, \tau_{2k+1} = (-1)^k$ , so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing  $M(n) = \begin{pmatrix} P(n) & -n^6 \\ 1 & 0 \end{pmatrix}$ , so that  $N_h(x) = 0$  iff a fixed matrix coefficient of the interval product  $M(x+h-1) \cdots M(x+1)$  vanishes), the conjectured  $\mathcal{E}_p(H) \ll H^{3/2}$  is analogous to a Rudnev-type point-plane incidence bound in  $\text{PGL}_2(\mathbb{F}_p)$ , subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

| $p$                            | 71   | 503  | 2003 | 5003 | 7919 | 20011 | 50021 | $10^5+3$ |
|--------------------------------|------|------|------|------|------|-------|-------|----------|
| $H = \lfloor \sqrt{p} \rfloor$ | 8    | 22   | 44   | 70   | 88   | 141   | 223   | 316      |
| $R_p(H)/H$                     | 1.75 | 1.59 | 1.36 | 1.50 | 1.75 | 1.63  | 1.56  | 1.46     |
| $\mathcal{E}_p(H)$             | 1    | 2    | 2    | 2    | 10   | 0     | 2     | 0        |
| $\mathcal{E}_p(H)/H^{3/2}$     | .044 | .019 | .007 | .003 | .012 | 0     | .001  | 0        |

The ratio  $R_p(H)/H$  clusters around  $3/2$  (the Chebotarev prediction from Remark 17), and  $\mathcal{E}_p(H)/H^{3/2} \rightarrow 0$ , consistent with  $\mathcal{E}_p(H) \ll H^{3/2}$ . The key arithmetic object is the shifted resultant  $S_{d,r} = \text{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$ : for  $p \nmid S_{d,r}$ , the gap polynomials  $N_d$  and  $N_r(\cdot+d)$  share no root over  $\mathbb{F}_p$ , so that gap pair contributes zero to  $\mathcal{E}_p$ . A Hadamard bound gives  $\log |S_{d,r}| \ll (d+r)^2 \log(d+r)$ , so the total number of prime divisors across all pairs  $(d, r)$  with  $d+r \leq H$  is  $O(H^4 \log H)$ . Averaging over the  $\sim X/\log X$  primes in  $(X, 2X]$  gives  $\mathcal{E}_p(H) \ll H^{1/2}$  for all but  $o(X/\log X)$  primes, provided  $H \leq X^{2/7-\varepsilon}$ —rigorous but far from the global target  $H = p-1$ .

A sharper route uses the bipartite incidence graph  $\Gamma_p(H)$ : left vertices are gap lengths  $d$ , right vertices are gap lengths  $r$ , with an edge when  $N_d$  and  $N_r(\cdot+d)$  share a root over  $\mathbb{F}_p$ . Each edge certifies a triple collision, so  $\mathcal{E}_p(H) = e(\Gamma_p(H))$ . A uniform fiber bound  $\deg_L(d) \leq C$  (the number of  $r$ -partners for each  $d$ ) and a uniform codegree bound (at most  $C$  common right neighbors for any two left vertices) would place  $\Gamma_p(H)$  in the regime of the Kővári–Sós–Turán theorem, giving  $\mathcal{E}_p(H) \ll H^{3/2}$ —the collision-energy scale needed for the mesoscopic target  $R_p(\sqrt{p}) = O(\sqrt{p})$ . The Dodgson identity (Remark 15) restricts local configurations in  $\Gamma_p$ , but by itself controls  $O(H)$  pairs out of  $O(H^2)$ ; upgrading it to a global codegree bound is the key missing step.

## 5.4 Computational evidence

**Proposition 28.** *For all 78,496 primes  $5 \leq p \leq 10^6$ :*

1.  $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$  for all but two primes; the two exceptions ( $p = 11$  and  $p = 3137$ ) have  $Z(p) = 1$  (non-ordinary primes, where  $p \mid a_p(f)$ ).  $\max Z(p) = 12$  at  $p = 159,977$ .
2. The exact histogram is:

| $Z(p)$                    | 0     | 1    | 2     | 4    | 6    | 8    | 10   | 12   |
|---------------------------|-------|------|-------|------|------|------|------|------|
| count                     | 47632 | 2    | 23730 | 6045 | 951  | 123  | 12   | 1    |
| fraction                  | .607  | .000 | .302  | .077 | .012 | .002 | .000 | .000 |
| $\text{Poi}(\frac{1}{2})$ | .607  | —    | .303  | .076 | .013 | .002 | .000 | .000 |

The mean  $Z(p) = 1.000$ , stable across all ranges ( $\chi^2 = 3.42$  on 4 d.f. against  $\text{Poisson}(1/2)$  for the pair count). The empirical factorial moments of  $K(p) = Z(p)/2$  over the 78,494 ordinary primes are:

| $r$                                     | 1     | 2     | 3     | 4     | 5     | 6     |
|-----------------------------------------|-------|-------|-------|-------|-------|-------|
| $\frac{1}{\pi_{\text{ord}}} \sum (K)_r$ | .4998 | .2490 | .1210 | .0605 | .0275 | .0092 |
| $2^{-r}$                                | .5000 | .2500 | .1250 | .0625 | .0312 | .0156 |

The first four moments match  $\text{Poisson}(1/2)$  to better than 3%; the higher-moment deviation is consistent with sampling variance ( $K \geq 5$  has only 13 primes in this range).

3. The symmetry  $b_j \equiv b_{p-1-j} \pmod{p}$  holds for all tested primes: zeros come in palindromic pairs  $\{j, p-1-j\}$ , so  $Z(p)$  is even (except at non-ordinary primes). The pair count  $Z(p)/2$  fits  $\text{Poisson}(1/2)$  to within sampling noise (table above). Proposition 18 proves  $\text{Poisson}(1/2)$  rigorously in the prime-aspect; the orbit-aspect fit (Remark 19) remains empirical.

*Remark 29* (Mesoscopic root count). At the scale  $H = \lfloor \sqrt{p} \rfloor$ , the total root count  $R_p(H) = \sum_{h=2}^H r_p(h)$  satisfies  $R_p(H)/H \approx 3/2$  for all tested primes, consistent with the Chebotarev prediction from Remark 17:

| $p$    | $H = \lfloor \sqrt{p} \rfloor$ | $R_p(H)$ | $R_p(H)/H$ |
|--------|--------------------------------|----------|------------|
| 503    | 22                             | 35       | 1.59       |
| 2003   | 44                             | 60       | 1.36       |
| 5003   | 70                             | 105      | 1.50       |
| 10007  | 100                            | 154      | 1.54       |
| 20011  | 141                            | 230      | 1.63       |
| 50021  | 223                            | 347      | 1.56       |
| 100003 | 316                            | 460      | 1.46       |

The ratio  $R_p(H)/H$  stabilizes near  $3/2$  (range 1.32–1.75 for  $p > 500$ ), supporting the mesoscopic conjecture in Remark 23.

## 5.5 Hypothesis Z

**Hypothesis 30** (Hypothesis Z). *There exists an absolute constant  $C$  such that  $Z(p) \leq C$  for all primes  $p$ .*

*Remark 31.* The analogous problem for the  $\zeta(2)$  Apéry numbers  $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$  has a fundamentally different structure. Stienstra–Beukers [20] showed that  $A_p$  is related to a CM modular form, yielding  $A_{(p-1)/2} \equiv 0 \pmod{p}$  for all  $p \equiv 3 \pmod{4}$  — a *positive density* of primes with guaranteed zeros. In contrast, the  $\zeta(3)$  Apéry numbers are governed by the non-CM newform  $f(z) = \eta(2z)^4 \eta(4z)^4 \in S_4(\Gamma_0(8))$  (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis  $\bar{Z}$  should hold for the  $\zeta(3)$  sequence, even though the pointwise bound  $Z(p) \leq C$  (Hypothesis Z) likely fails.

*Remark 32.* Even the single-row slice  $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$  is connected to the non-ordinary primes of  $f$  via the Beukers congruence  $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$  [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [14]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is  $\sim C_f/p$  (the Sato–Tate distribution on the normalized coefficients  $a_p/p^{3/2}$  admits  $\sim 4p^{1/2}$  multiples of  $p$  in  $[-2p^{3/2}, 2p^{3/2}]$ , each contributing mass  $\sim p^{-3/2}$ ). The expected count up to  $x$  is thus  $C_f \log \log x$  with  $C_f \approx 1$ . Finding 2 non-ordinary primes ( $p = 11, 3137$ ) below  $10^6$  is consistent with  $\log \log(10^6) \approx 2.9$ . This is not a standard Lang–Trotter problem: the target set  $\{a : p \mid a\}$  varies with  $p$ , and no analogue of Elkies’ theorem (infinitely many supersingular primes) is known for weight  $\geq 4$ .

*Remark 33* (Evidence against Hypothesis Z). The  $2 \cdot \text{Poisson}(1/2)$  distribution (Remark 9 below) has unbounded support: extreme-value theory predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$ . The current record  $Z(p) = 12$  at  $p = 159,977$  (with record progression 2, 4, 6, 8, 10, 12) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with  $Z(p) = 14$  should appear near  $p \approx 2 \times 10^7$  (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to  $p = 5 \times 10^7$  would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

**Hypothesis 34** (Hypothesis  $\bar{Z}$ ).  $\sum_{p \leq x} Z(p) \ll \pi(x)$ , *equivalently, the average of  $Z(p)$  over primes  $p \leq x$  is bounded.*

*Remark 35.* Hypothesis  $\bar{Z}$  is strictly weaker than Hypothesis  $Z$  and consistent with  $Z(p) \sim 2 \cdot \text{Poisson}(1/2)$  (which gives  $\mathbb{E}[Z(p)] = 1$ , independent of  $p$ ). It suffices for the conditional bound  $\log G_n = O(\sqrt{n})$  (Theorem 55).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

**Hypothesis 36** (Poisson conjecture). *For every good prime  $p \geq 5$ , put*

$$\nu_p := \mathbf{1}_{\{p|a_p(f)\}}, \quad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

*where  $f$  is the weight-4 newform associated to the Apéry differential operator. Thus  $K_*(p)$  counts noncentral reflected pairs of zeros of  $(b_j \bmod p)_{0 \leq j < p}$ . For every fixed  $r \geq 1$ ,*

$$\frac{1}{\pi(X)} \sum_{p \leq X} \binom{K_*(p)}{r} \rightarrow \frac{1}{2^r r!}.$$

*On ordinary primes ( $\nu_p = 0$ ), this reduces to  $K(p) = Z(p)/2 \rightarrow \text{Poisson}(1/2)$ , consistent with the data in Proposition 28.*

*Remark 37.* The Poisson conjecture implies Hypothesis  $\bar{Z}$ . Taking  $r = 1$  gives  $\sum_{p \leq X} K_*(p) = (\frac{1}{2} + o(1)) \pi(X)$ . Since  $Z(p) = 2K_*(p) + \nu_p$  and  $0 \leq \nu_p \leq 1$ ,

$$\sum_{p \leq X} Z(p) = 2 \sum_{p \leq X} K_*(p) + \sum_{p \leq X} \nu_p \leq (2 + o(1)) \pi(X),$$

which is Hypothesis  $\bar{Z}$ . A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

*Remark 38* (Computational evidence for  $\bar{Z}$ ). Over dyadic blocks of primes,  $(\sum_{P < p \leq 2P} Z(p))/|\{P < p \leq 2P\}|$  ranges from 0.86 to 1.07 for  $P = 2^k$ ,  $3 \leq k \leq 12$  (666 primes up to 4999), with overall average  $\sum Z(p)/\pi = 0.91$ . The Poisson prediction is 1.

## 6 The denominator connection lemma

**Lemma 39.** *Let  $p \geq 7$  be prime with  $n/2 < p \leq n$ , and set  $r = n - p$ . Then  $v_p(G_n) = 0$  if and only if  $p \nmid b_r$ .*

*Proof.* Since  $p > n/2$ , the only multiple of  $p$  in  $\{1, \dots, n\}$  is  $p$  itself, so  $v_p(d_n) = 3$ .

**Step 1.**  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $v_p(G_n) = 0$  requires  $v_p(d_n a_n) = 0$ , i.e.,  $v_p(a_n) = -3$ .

**Step 2 (variation of parameters).** From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^n \frac{6}{m^3 b_m b_{m-1}}. \quad (12)$$

For  $p > n/2$ , the only  $m \in \{1, \dots, n\}$  with  $p \mid m$  is  $m = p$ .

**Step 3.** By Theorem 4,  $b_p \equiv b_0 b_1 = 5 \pmod{p}$  (since  $p$  has base- $p$  digits  $(0, 1)$ ), and by the symmetry  $b_j \equiv b_{p-1-j}$ ,  $b_{p-1} \equiv b_0 = 1 \pmod{p}$ . Since  $p \geq 7$ , both are nonzero mod  $p$ .

The  $m = p$  term in (12) has  $v_p = -3$ ; all other terms have  $v_p \geq 0$ . Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p b_{p-1}} \equiv \frac{6}{5} \pmod{p}.$$

Using  $b_n \equiv 5b_r \pmod{p}$  (Theorem 4, since  $n = 1 \cdot p + r$ ):

$$p^3 a_n \equiv \frac{6}{5} \cdot 5b_r = 6b_r \pmod{p}.$$

Thus  $v_p(a_n) = -3$  iff  $p \nmid b_r$ . □

**Verification.** For all  $n \leq 300$  and primes  $p \in (n/2, n]$  with  $p \geq 7$ :  $v_p(G_n) = 0 \iff p \nmid b_{n-p}$  confirmed in 2101/2101 cases (100%).

**Corollary 40** (Lucas reduction). *For  $p \geq 7$  and  $p \in (n/2, n]$ :  $v_p(G_n) \geq 1$  iff  $p \mid b_{n-p}$ . By the Lucas congruence  $b_n \equiv b_1 b_{n-p} = 5b_{n-p} \pmod{p}$ , this is equivalent (for  $p \neq 5$ ) to  $p \mid b_n$ . Thus  $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$  equals  $\omega_{(n/2, n]}(b_n) + O(1)$ , the number of distinct prime factors of  $b_n$  lying between  $n/2$  and  $n$ , up to an absolute constant.*

The full conjecture  $G_n = e^{o(n)}$  for all  $n$  is therefore equivalent to the *top-half radical estimate*:

$$\log \text{rad}_{(n/2, n]}(b_n) = o(n). \quad (13)$$

Since  $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52n$ , the height bound gives  $\omega_{(n/2, n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$ , precisely one  $o(1)$  factor short of the target. Computation for  $n \leq 200,000$  gives  $B(n) \leq 3$  and  $B(n) = 0$  for 93.7% of values (see Remark 41 for the Poisson analysis).

*Remark 41* (Random-model prediction). Under a random-integer model,  $\Pr(p \mid b_n) \approx 1/p$  for each prime  $p$ , so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\left(\frac{\log n}{\log(n/2)}\right) + O(1/\log n) \approx \frac{\log 2}{\log n} \rightarrow 0.$$

Computation for  $n \leq 200,000$  confirms:  $\mathbb{E}[B] \approx 0.065$ ,  $\max B(n) = 3$ , and the histogram matches  $\text{Poisson}(\log 2 / \log n)$  to high accuracy:  $B(n) = 0$  for 93.7%,  $B(n) = 1$  for 6.0%,  $B(n) = 2$  for 0.20%,  $B(n) = 3$  for 0.006% (the Poisson predictions with  $\lambda = 0.065$  are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio  $\text{Var}(B)/\mathbb{E}[B] = 1.003$  is the exact Poisson signature. By CRT-based approximate independence across primes,  $B(n)$  should follow  $\text{Poisson}(\log 2 / \log n)$  with  $\lambda \rightarrow 0$ . Extreme-value theory then predicts  $\max_{n \leq N} B(n) = O(\log N / \log \log N)$ —trivially  $o(n/\log n)$ —so the random model *strongly* predicts the full conjecture. The obstruction is proving that  $b_n$  behaves “randomly” with respect to its prime factors in  $(n/2, n]$ , despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

## 7 The unconditional density-one theorem

**Theorem 42.** *For a set of positive integers  $n$  of natural density 1,  $G_n = e^{o(n)}$ .*

*Proof.* Fix  $N$  large and consider  $n \in (N, 2N]$ . Write  $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$  for every  $n$ . Each such prime contributes  $v_p(G_n) \log p \leq 3 \log n$  (since  $v_p(d_n) = 3$  for  $p > \sqrt{n}$ ), so  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ . It suffices to show  $B(n) = o(n/\log n)$  for all but  $o(N)$  values of  $n \in (N, 2N]$ .

For  $p > \sqrt{n}$ , write  $n = qp + r$  with  $1 \leq q \leq \sqrt{2N}$  and  $0 \leq r < p$ . By Lemma 47(iii),  $v_p(G_n) \geq 1$  requires  $D_q b_r \equiv 0 \pmod{p}$ . By Lemma 48, leading-digit bad primes ( $p \mid b_q$ ,  $p \nmid b_r$ ) have  $v_p(G_n) = 0$  and do not contribute to  $B(n)$ . The remaining contributions come from *lower-digit bad primes* ( $r \in \mathcal{Z}_p$ ) and *companion-block primes* ( $D_q \equiv 0 \pmod{p}$ ,  $p \nmid b_r$ ).

**Lower-digit incidence.** For fixed  $p \in (\sqrt{N}, 2N]$ , an interval of  $N$  consecutive integers contains at most  $N \cdot Z(p)/p + Z(p)$  values of  $n$  whose residue  $r = n \bmod p$  lies in  $\mathcal{Z}_p$ . Set  $\rho_p = Z(p)/p$ . The total lower-digit incidence is

$$R_N \leq \sum_{\sqrt{N} < p \leq 2N} (N\rho_p + Z(p)).$$

By Proposition 9,  $\rho_p \rightarrow 0$ . Since  $\pi(2N) = O(N/\log N)$ :

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \leq 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence  $R_N = o(N^2/\log N)$ .

**Companion-block incidence.** By Proposition 51, the companion-block contribution to  $\log G_n$  is  $O(n^{2/3}) = o(n)$  at every index. Hence companion-block primes do not affect the density-one statement, and we set  $P_N = 0$  in the counting argument.

**Conclusion.** Write  $\sum_{n \in (N, 2N]} B(n) \leq R_N + P_N = o(N^2/\log N)$ . For any fixed  $\delta > 0$ , Markov's inequality gives

$$\#\{n \in (N, 2N] : B(n) > \delta n / \log n\} \leq \frac{o(N^2/\log N)}{\delta N / \log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so  $B(n) = o(n/\log n)$  for density 1. For all such  $n$ ,  $\log G_n = O(\sqrt{n}) + 3B(n) \log n = O(\sqrt{n}) + o(n) = o(n)$ .  $\square$

**Corollary 43** (Quantitative exceptional set). *For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{2/3}).$$

*Proof.* We make the density-one argument quantitative using the explicit bound  $Z(p) = O(p^{2/3})$  from Proposition 9. For  $n \in (N, 2N]$ ,  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$ , so  $\log G_n > \varepsilon n$  requires  $B(n) > \varepsilon n / (7 \log N)$  for  $N$  large.

Partial summation with  $Z(p) \leq Cp^{2/3}$  gives

$$\sum_{\sqrt{N} < p \leq 2N} \frac{Z(p)}{p} \leq C \sum_{p \leq 2N} p^{-1/3} = \frac{3}{2} C \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies  $R_N = N \sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$  (the second sum,  $\sum Z(p) \leq C \sum p^{2/3}$ , contributes the same order). The companion-block incidence  $P_N = O(N^{3/2})$  is dominated by  $R_N$ . Hence  $\sum_{n \in (N, 2N]} B(n) = O(N^{5/3}/\log N)$ .

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon n / (7 \log N)\} \leq \frac{O(N^{5/3}/\log N)}{\varepsilon N / (7 \log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals  $(N/2^{j+1}, N/2^j]$  gives the global bound, since  $\sum_{j \geq 0} (N/2^j)^{2/3} = O(N^{2/3})$ .  $\square$

In particular, the exceptional set has finite harmonic weight:

**Corollary 44** (Finite harmonic weight). *For every  $\varepsilon > 0$ ,*

$$\sum_{\substack{n \geq 1 \\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

*Proof.* The contribution from  $(2^k, 2^{k+1}]$  is  $O(2^{-k/3})$  by Corollary 43, and the geometric series converges.  $\square$

*Remark 45.* The exponent  $2/3$  in the exceptional set is inherited from the lower-digit incidence  $R_N$ , which uses  $Z(p) = O(p^{2/3})$ . More generally, if  $Z_p(I) \leq C|I|^\alpha$  uniformly for all primes and sub-intervals (as Corollary 10 provides with  $\alpha = 2/3$ ), then  $R_N = O(N^{1+\alpha}/\log N)$  and  $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$ . The exceptional set is therefore  $O(\max(N^\alpha, N^{1/(3-\alpha)})/\varepsilon)$ , with the crossover at  $\alpha = (3 - \sqrt{5})/2 \approx 0.382$ . Under Hypothesis  $\bar{Z}$ , the dyadic bound  $D(P, Q) \leq \sum Z(p) = O(P/\log P)$  is equivalent to  $\alpha = 0$  in this formula, giving the exceptional set  $O(N^{1/3}/\varepsilon)$  (Corollary 57). Theorem 55 shows the matching density-one bound  $\log G_n = O(\sqrt{n})$ , where  $O(\sqrt{n})$  is now the small-prime floor, not the leading-digit contribution.

## 8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

**Lemma 46** (Consecutive coprimality). *Let  $p > 5$  be prime and  $n \geq p$ . If  $p \mid b_{n \bmod p}$  and  $p \mid b_{(n+1) \bmod p}$ , then  $p \leq n/2$ . Equivalently, no prime  $p > n/2$  with  $p > 5$  can be lower-digit bad for both  $n$  and  $n + 1$ .*

*Proof.* Since  $p > n/2$ , both  $n$  and  $n + 1$  have base- $p$  representations  $n = p + r$ ,  $n + 1 = p + (r + 1)$  with  $0 \leq r < r + 1 < p$  (provided  $r < p - 1$ ; the case  $r = p - 1$  gives  $n + 1 = 2p$  and  $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$  for  $p > 73$ , a finite check).

Suppose  $p \mid b_r$  and  $p \mid b_{r+1}$  with  $r, r + 1 < p$ . The Apéry recurrence gives

$$(k + 1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since  $k + 1 < p$  implies  $(k + 1)^3$  is invertible mod  $p$ , the conditions  $p \mid b_k$  and  $p \mid b_{k+1}$  force  $p \mid b_{k-1}$ . Iterating downward from  $k = r$  gives  $p \mid b_j$  for all  $0 \leq j \leq r + 1$ , contradicting  $b_0 = 1$ .  $\square$

A prime that divides  $b_n$  only through a leading base- $p$  digit contributes nothing to  $G_n$ .

**Lemma 47** (Block system for  $p^3 a_n \bmod p$ ). *Let  $p \geq 7$ ,  $p \leq n < p^2$ , and write  $n = qp + r$  with  $1 \leq q < p$  and  $0 \leq r < p$ . Define  $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod{p}$  and  $D_q = \Delta_{q,0}$ . Then:*

- (i)  $\Delta_{q,r} \equiv D_q b_r \pmod{p}$  for every  $0 \leq r < p$ ;
- (ii)  $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod{p}$ ;
- (iii)  $p \mid G_n$  if and only if  $D_q b_r \equiv 0 \pmod{p}$ .

*Proof.* For  $p > \sqrt{n}$ ,  $v_p(\text{lcm}(1, \dots, n)) = 1$ , so  $v_p(d_n) = 3$ . Since  $d_n a_n \in \mathbf{Z}$ ,  $p^3 a_n \in \mathbf{Z}_p$ , and  $v_p(d_n a_n) = v_p(p^3 a_n)$ . Meanwhile  $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$ , so  $p \mid G_n \Leftrightarrow p \mid p^3 a_n$ .

(i) Multiply the Apéry recurrence at index  $qp + r$  by  $p^3$  and reduce modulo  $p$ . Since  $qp + r \equiv r \pmod{p}$ , the sequence  $r \mapsto \Delta_{q,r}$  satisfies the same recurrence  $(r + 1)^3 \Delta_{q,r+1} = P(r) \Delta_{q,r} - r^3 \Delta_{q,r-1}$

as  $r \mapsto D_q b_r$ , with matching initial values  $\Delta_{q,0} = D_q$  and  $\Delta_{q,1} = 5D_q = D_q b_1$ . Since  $(r+1)^3$  is invertible mod  $p$  for  $0 \leq r \leq p-2$ , uniqueness gives the identity.

(ii) Multiply the Wronskian  $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$  at  $n = qp$  by  $p^3$  and reduce mod  $p$ , using  $b_{qp} \equiv b_q$  and  $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$  (Lucas), and  $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$  (part (i)).

(iii) Immediate from (i) and the observation above.  $\square$

**Lemma 48** (Leading-digit vanishing). *Let  $p \geq 7$ ,  $p \leq n < p^2$ ,  $n = qp + r$  with  $q \geq 1$ . If  $p \mid b_q$  and  $p \nmid b_r$ , then  $v_p(G_n) = 0$ .*

*Proof.* Since consecutive Apéry numbers below the  $p$ -th cannot both vanish mod  $p$  (the recurrence propagates to  $b_0 = 1$ ),  $p \nmid b_{q-1}$ . Part (ii) of Lemma 47 gives  $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod{p}$ , so  $D_q \not\equiv 0$ . By part (iii),  $p \mid G_n$  iff  $D_q b_r \equiv 0$ , but  $D_q \not\equiv 0$  and  $p \nmid b_r$ , so  $D_q b_r \not\equiv 0$  and  $v_p(G_n) = 0$ .  $\square$

*Remark 49* (Block constants are Apéry numbers). The block constant  $D_q$  of Lemma 47 equals  $a_q \pmod{p}$ , where  $a_q$  is the Apéry rational companion. Indeed,  $a_q$  satisfies the same Wronskian recurrence  $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$  with  $a_0 = 0$ ,  $a_1 = 6$ , so  $D_q = a_q$  by uniqueness. Consequently,  $D_q \equiv 0 \pmod{p}$  iff  $p$  divides the numerator of  $a_q$  (in lowest terms) and  $p$  does not divide its denominator.

Since  $|a_q| \leq C(1 + \sqrt{2})^{4q}$  and  $\text{denom}(a_q) \mid \text{lcm}(1, \dots, q)^3$ , the numerator of  $a_q$  has  $O(q)$  prime divisors. In particular, for fixed  $q$ , at most  $O(q)$  primes  $p$  can be companion-block bad at quotient  $q$ . Computation through  $n = 2000$  confirms that the companion-block count  $\#\{p > \sqrt{n} : p \mid \text{num}(a_{\lfloor n/p \rfloor})\}$  is at most 3 for every  $n$  in this range.

**Proposition 50** (Reflection of the companion solution). *Let  $p \geq 5$  be prime. Every  $p$ -integral solution  $(u_n)$  of the Apéry recurrence (1) on  $\{0, \dots, p-1\}$  is palindromic modulo  $p$ :*

$$u_{p-1-j} \equiv u_j \pmod{p} \quad (0 \leq j \leq p-1).$$

*In particular, for the companion solution  $a_0 = 0$ ,  $a_1 = 6$ ,*

$$a_{p-1-j} \equiv a_j \pmod{p}, \quad a_{p-1} \equiv 0 \pmod{p}.$$

*Proof.* Let  $R$  denote the reversal operator,  $(Ru)_j = u_{p-1-j}$ . The coefficient polynomial  $P(n) = 34n^3 + 51n^2 + 27n + 5$  satisfies  $P(-1-x) = -P(x)$ , so substituting  $n \mapsto p-2-n$  in the recurrence shows that  $R$  maps  $p$ -integral solutions to  $p$ -integral solutions modulo  $p$ .

The integer Apéry solution  $(b_n)$  satisfies  $b_{p-1} \equiv 1$  and  $b_{p-2} \equiv 5 \pmod{p}$ : in the binomial sum  $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ , only the  $k=0$  term survives modulo  $p$  for  $n = p-1$ , while for  $n = p-2$  only  $k=0, 1$  survive, contributing  $1+4=5$ . Since  $Rb$  and  $b$  have the same initial values  $(b_0, b_1) = (1, 5)$ , uniqueness gives  $Rb = b$ .

For two solutions  $x, y$ , the discrete Wronskian  $W_n(x, y) = x_n y_{n-1} - x_{n-1} y_n$  satisfies  $(n+1)^3 W_{n+1} = n^3 W_n$ , so  $W_n(x, y) = C(x, y)/n^3$ . Using  $Rb = b$ :

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence  $W_n(Ru - u, b) = 0$ , so  $Ru - u = cb$  for some  $c \in \mathbf{F}_p$ . Applying  $R$ : on one hand  $R(Ru - u) = -(Ru - u)$ , on the other  $R(cb) = cb$ . Since  $p$  is odd,  $c = 0$  and  $Ru = u$ .  $\square$

**Proposition 51** (Companion-block height bound). *Assume  $D_q \equiv a_q \pmod{p}$ . For every  $n \geq 1$ , the total logarithmic weight of companion-block primes is at most  $O(n^{2/3})$ :*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \text{num}(a_{\lfloor n/p \rfloor})}} \log p \leq C n^{2/3}$$

for an absolute constant  $C$ . In particular, companion-block primes contribute  $o(n)$  to  $\log G_n$  at every index.

*Proof.* Write  $a_q = A_q/C_q$  in lowest terms. From the telescoping identity  $a_q = b_q \sum_{k=1}^q 6/(k^3 b_k b_{k-1})$  and  $b_k \geq 1$ , we obtain  $|A_q| \leq |a_q| \cdot C_q \leq C_0(1 + \sqrt{2})^{4q}$ , hence each numerator  $A_q$  has at most  $O(q)$  prime divisors.

For a fixed  $n$ , set  $Q = \lfloor n^{1/3} \rfloor$ . A companion-block prime  $p > \sqrt{n}$  has quotient  $q = \lfloor n/p \rfloor < \sqrt{n}$ .

*Small quotients* ( $q \leq Q$ ): for each  $q$ , the primes dividing  $A_q$  contribute  $\sum \log p \leq \log |A_q| = O(q)$ . Summing over  $q \leq Q$ :  $\sum_{q=1}^Q O(q) = O(Q^2) = O(n^{2/3})$ .

*Large quotients* ( $q > Q$ ): each such prime satisfies  $p < n/Q = n^{2/3}$ . Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum  $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$ .  $\square$

The first-moment bound in Corollary 43 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

**Lemma 52** (Common bad-prime codegree). *Let  $m, n \in (N, 2N]$  with  $h = n - m \geq 1$ , and let  $P_0 \geq 1$ . The number of primes  $p > P_0$  that are simultaneously lower-digit bad for  $m$  and for  $n$  (i.e.,  $p \mid b_{m \bmod p}$  and  $p \mid b_{n \bmod p}$ ) is at most  $O(h \log N / \log P_0)$ . In particular, if  $P_0 \geq N^\delta$  for any fixed  $\delta > 0$ , the codegree is  $O(h)$ .*

*Proof.* Since  $p > N/2 \geq h$ , the residues  $m \bmod p$  and  $n \bmod p$  lie in  $\{0, \dots, p-1\}$  and differ by exactly  $h$ . By the gap implication (Lemma 7), if  $p \mid b_{m \bmod p}$  and  $p \mid b_{n \bmod p}$ , then  $p \mid N_h(m \bmod p)$ . The polynomial periodicity of  $N_h$  gives  $N_h(m \bmod p) \equiv N_h(m) \pmod{p}$ , so  $p \mid N_h(m)$ .

The gap polynomial has degree  $3(h-1)$  in  $m$  (the tridiagonal determinant representation gives exactly this), so for  $m \in (N, 2N]$ ,  $|N_h(m)| \leq (CN^3)^h$ . Since  $N_h(m) > 0$  (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding  $P_0$  is at most  $3h \log(CN) / \log P_0$ .  $\square$

**Theorem 53** (Polylogarithmic exceptional set). *For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon((\log N)^2).$$

*Proof.* Fix  $\varepsilon > 0$  and consider  $n \in (N, 2N]$  with  $\log G_n > \varepsilon n$ . By Proposition 3, the small-prime contribution is  $O(\sqrt{n})$ . Each prime  $p > \sqrt{n}$  satisfies  $v_p(G_n) \leq 3$  (since  $v_p(d_n) = 3$ ). By Lemma 48, leading-digit bad primes ( $p \mid b_{\lfloor n/p \rfloor}$  with  $n \bmod p \notin \mathcal{Z}_p$ ) satisfy  $v_p(G_n) = 0$ . The remaining primes with  $v_p(G_n) \geq 1$  are the lower-digit bad primes ( $n \bmod p \in \mathcal{Z}_p$ ) and the *companion-block primes* ( $D_q \equiv 0 \pmod{p}$  with  $p \nmid b_r$ , cf. Lemma 47). By Proposition 51, the total logarithmic weight of companion-block primes is  $O(n^{2/3})$  at every index, hence  $o(n)$ . Hence  $\log G_n \leq O(\sqrt{n}) + 3t(n) \log n$ , and  $\log G_n > \varepsilon n$  forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{4 \log n} \quad (14)$$

for  $N$  sufficiently large.

Partition  $(N, 2N]$  into intervals  $J$  of length  $Y = c_1 \varepsilon^2 N / \log N$ , where  $c_1 > 0$  is a small constant to be chosen. Write  $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$  and  $M = |\mathcal{E}_J|$ .

For each prime  $p \in (\sqrt{N}, 2N]$ , let  $d_p$  count the elements of  $\mathcal{E}_J$  for which  $p$  is lower-digit bad. By (14),

$$I := \sum_p d_p \geq \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size  $L = \pi(2N) \leq 4N/\log N$ .

**Upper bound on pair incidence.** For  $m, n \in \mathcal{E}_J$  with  $m < n$ , Lemma 52 gives at most  $C(n - m)$  common lower-digit bad primes (taking  $P_0 = N^\delta$  in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \leq C \sum_{\substack{m < n \\ m, n \in \mathcal{E}_J}} (n - m) \leq CYM^2.$$

**Lower bound on pair incidence.** By Cauchy–Schwarz applied to the  $d_p$ ,

$$\sum_p \binom{d_p}{2} \geq \frac{1}{2} \left( \frac{I^2}{L} - I \right) \geq \frac{1}{2} \left( \frac{\varepsilon^2 NM^2}{64 \log N} - I \right).$$

Combining gives  $CYM^2 \geq \varepsilon^2 NM^2/(128 \log N) - I/2$ . Since  $I \leq LM \leq 4NM/\log N$  and  $Y = c_1 \varepsilon^2 N/\log N$ , for  $c_1$  chosen smaller than  $1/(128C)$ , the  $YM^2$  term cannot absorb the quadratic lower bound once  $M > C'/\varepsilon^2$ . Hence  $M = O(1/\varepsilon^2)$ .

There are  $N/Y = O(\log N/\varepsilon^2)$  intervals. The total on  $(N, 2N]$  is therefore  $O(\log N/\varepsilon^4)$ . Summing over  $O(\log N)$  dyadic blocks  $(N/2^{j+1}, N/2^j]$  gives the result.  $\square$

Theorem 53 supersedes Corollary 43. The proof’s critical-window estimate—every interval of length  $\asymp \varepsilon^2 N/\log N$  in  $(N, 2N]$  contains  $O_\varepsilon(1)$  exceptional integers—directly gives the uniform local bound

$$\sup_{M \geq 0} \#\{M < n \leq M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval  $[M, M + L]$  with  $N = M$  and  $Y = c\varepsilon^2 M/\log M$ . Dividing by  $L$  and letting  $L \rightarrow \infty$  yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 44.

## 9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from  $O(N^{3/2})$  to  $O(N^{10/7}/\log N)$  via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

**Proposition 54** (Dyadic leading-digit bound). *The leading-digit incidence  $L_N$  (counting pairs  $(n, p)$  with  $n \in (N, 2N]$ ,  $p > \sqrt{n}$  prime, and  $p \mid b_{\lfloor n/p \rfloor}$ ) satisfies*

$$L_N = O\left(\frac{N^{10/7}}{\log N}\right).$$

*Proof.* For dyadic blocks  $P = 2^j$ ,  $Q = 2^k$  with  $Q < P$  and  $PQ \in [N/4, 4N]$ , define

$$D(P, Q) = \#\{(p, q) : P < p \leq 2P, Q < q \leq 2Q, p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime  $p \in (P, 2P]$  has at most  $O(Q^{2/3})$  zeros of  $b_j$  in  $(Q, 2Q]$ . Summing over  $\pi(2P)$  primes gives  $D(P, Q) = O(PQ^{2/3}/\log P)$ .

- *Divisor counting:* Since  $\log b_q \leq (q+1) \log 64$ , the number of prime factors of  $b_q$  exceeding  $P$  is at most  $\log b_q / \log P = O(q / \log P)$ . Summing over  $q \in (Q, 2Q]$  gives  $D(P, Q) \leq \sum_q O(q / \log P) = O(Q^2 / \log P)$ .

Hence  $D(P, Q) \leq \min(P Q^{2/3}, Q^2) / \log P$ .

Each pair  $(p, q)$  in  $D(P, Q)$  contributes to at most  $p \leq 2P$  values of  $n$  with  $\lfloor n/p \rfloor = q$ , so the block's contribution to  $L_N$  is at most  $D(P, Q) \cdot 2P \leq 2 \min(P^{4/3} N^{2/3}, N^2/P) / \log P$  (using  $Q = \Theta(N/P)$ ). The crossover is at  $P_0 = N^{4/7}$ . Summing  $O(\log N)$  dyadic blocks  $P \in [\sqrt{N}, 2N]$ :

$$L_N \leq \sum_{P \leq N^{4/7}} \frac{2P^{4/3} N^{2/3}}{\log P} + \sum_{P > N^{4/7}} \frac{2N^2}{P \log P} = O\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the  $P = N^{4/7}$  boundary term.  $\square$

**Theorem 55.** *Assume Hypothesis  $\bar{Z}$ . Then for a set of positive integers  $n$  of natural density 1,*

$$\log G_n = O(\sqrt{n}).$$

*Proof.* Fix  $N$  and  $\omega(N) \rightarrow \infty$  arbitrarily slowly. As in Theorem 42, decompose  $\log G_n = \sum_p v_p(G_n) \log p$ . The small-prime contribution is  $O(\sqrt{n})$ . Split  $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$ , where  $B_{\text{low}}$  counts bad primes  $p > \sqrt{n}$  with  $n \bmod p \in \mathcal{Z}_p$  and  $B_{\text{lead}}$  counts those with  $\lfloor n/p \rfloor \in \mathcal{Z}_p$ .

**Lower digit.** Under Hypothesis  $\bar{Z}$ , set  $M(t) = \sum_{p \leq t} Z(p) \ll t / \log t$ . Abel summation gives  $\sum_{\sqrt{N} < p \leq 2N} Z(p)/p = O(1)$ , so  $R_N = \sum B_{\text{low}}(n) = O(N)$ . By Markov,  $B_{\text{low}}(n) \leq \omega(N)$  for all but  $O(N/\omega(N)) = o(N)$  values.

**Leading digit.** As in the proof of Corollary 57, Hypothesis  $\bar{Z}$  gives  $L_N = O(N^{4/3} / \log N)$ . By Markov with threshold  $\omega(N) N^{1/3} / \log N$ :

$$\#\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N) N^{1/3} / \log N\} \leq \frac{O(N^{4/3} / \log N)}{\omega(N) N^{1/3} / \log N} = O(N / \omega(N)) = o(N).$$

**Conclusion.** For all remaining  $n$ :

$$\log G_n \leq O(\sqrt{n}) + 3(\omega(N) + \omega(N) N^{1/3} / \log N) \log n.$$

Since  $\omega(N) \cdot N^{1/3} \ll N^{1/2}$  for any  $\omega(N) \rightarrow \infty$  slowly enough ( $\omega \leq N^{1/6}$  suffices), the second term is  $O(N^{1/2}) = O(\sqrt{n})$ .  $\square$

*Remark 56* (The leading-digit bottleneck). The gap between the conditional bound  $O(\sqrt{n})$  (Theorem 55) and the conjectured  $O(\log n)$  comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of  $O(\sqrt{n})$ . Under Hypothesis  $\bar{Z}$ , the big-prime contribution is now absorbed into this floor, thanks to Proposition 54.

Heuristically,  $B_{\text{lead}}(n) = O(1)$  because the leading digit  $\lfloor n/p \rfloor$  lands in  $\mathcal{Z}_p$  with probability  $\leq C/p$  for each prime, giving expected  $B_{\text{lead}} = \sum C/p = O(1)$ . Making this rigorous would require  $D(P, Q) \ll Q / \log P$  (the random-divisibility prediction), improving the proven  $D(P, Q) \leq \min(P Q^{2/3}, Q^2) / \log P$ . The small-prime bound  $O(\sqrt{n})$  itself would need a finer analysis of  $v_p(G_n)$  for small primes.

**Corollary 57** (Conditional exceptional set). *Assume Hypothesis  $\bar{Z}$ . For every  $\varepsilon > 0$ ,*

$$\#\{n \leq N : \log G_n > \varepsilon n\} = O_\varepsilon(N^{1/3}).$$

*Proof.* Under Hypothesis  $\bar{Z}$ ,  $R_N = O(N)$ . For the leading-digit incidence, Hypothesis  $\bar{Z}$  also strengthens the dyadic bound: since each prime  $p \in (P, 2P]$  contributes at most  $Z(p)$  zeros to any sub-interval,  $D(P, Q) \leq \sum_{P < p \leq 2P} Z(p) = O(P/\log P)$  (using  $\sum_{p \leq x} Z(p) = O(\pi(x))$ ). The crossover with the divisor bound  $D(P, Q) \leq Q^2/\log P$  shifts from  $P_0 = N^{4/7}$  to  $P_0 = N^{2/3}$ , giving  $L_N = O(N^{4/3}/\log N)$ . The threshold  $B(n) > \varepsilon N/(7 \log N)$  is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7 \log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over  $(N/2^{j+1}, N/2^j]$  gives the global bound.  $\square$

Under Hypothesis  $\bar{Z}$ , the exceptional set is not only sparse globally but cannot cluster in any interval:

**Corollary 58** (Upper Banach density zero). *Assume Hypothesis  $\bar{Z}$ . For every  $\varepsilon > 0$ , the set  $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$  has upper Banach density zero:*

$$\lim_{L \rightarrow \infty} \sup_{M \geq 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

*Proof.* For  $I = (M, M+L]$  with  $M > 2L$ , every prime  $p > n/2$  with  $n \in I$  lies in  $(M/2, M+L]$ . Under Hypothesis  $\bar{Z}$ , each such prime contributes at most  $Z(p)$  candidate bad integers to  $I$ , giving  $\sum_{n \in I} B(n) \leq \sum_{M/2 < p \leq M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$ . For  $n \in I \cap E_\varepsilon$ , the bound  $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$  and  $\log G_n > \varepsilon n$  force  $B(n) \gg \varepsilon M/\log M$  for  $M$  large. Therefore  $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$ . The case  $M \leq 2L$  follows similarly by covering  $(0, 3L]$  with  $O(\log L)$  dyadic intervals.  $\square$

**Corollary 59.** *The condition  $Z(p) = o(p)$  is necessary for the conjecture: if  $Z(p) \geq \varepsilon p$  for a positive proportion of primes, double-counting gives  $\log G_n \geq c\varepsilon n$  for a positive proportion of  $n$ . Conversely,  $Z(p) = o(p)$  implies  $G_n = e^{o(n)}$  for density 1 (Theorem 42).*

*Proof.* For each prime  $p \in (n/2, n]$  with  $Z(p) \geq \varepsilon p$ , Lemma 39 gives  $v_p(G_n) \geq 1$  whenever  $n-p \in \mathcal{Z}_p$ . Suppose a proportion  $\delta > 0$  of primes satisfy  $Z(p) \geq \varepsilon p$ . Summing over  $n \in (N, 2N]$  and bad primes  $p \in (n/2, n]$ , the total incidence count is  $\geq \delta \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$ . Dividing by  $N$  and applying Markov's inequality: a proportion  $\geq \varepsilon\delta/4$  of integers  $n$  have  $B(n) \geq \varepsilon\delta N/(4 \log N)$ , giving  $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta n$ .  $\square$

*Remark 60* (The pointwise gap). The full conjecture  $G_n = e^{o(n)}$  for every  $n$  is strictly stronger than the density-one statement. By Corollary 40, it is equivalent to the top-half radical estimate (13). The height bound gives  $B(n) = O(n/\log n)$ , but for an *adversarial* family  $(Z_p)$  with  $|Z_p| \leq 1$ , one may choose  $Z_p = \{N-p\}$  for primes in  $(N/2, N]$ , producing  $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$ . Thus the needed pointwise bound  $B(n) = o(n/\log n)$  requires arithmetic information specific to the Apéry sequence, not just the cardinalities  $Z(p)$ .

By Corollary 40, every bad prime  $p > n/2$  satisfies  $p \mid b_n$ . Hence the conjecture implies  $\log \text{rad}_{(n/2, n]}(b_n) = o(n)$ : the large prime radical of  $b_n$  must be subexponential, even though  $|b_n| = e^{\Theta(n)}$ .

The Poisson model predicts that  $B(n) \geq 1$  occurs infinitely often:  $\mathbb{P}(B(n) \geq 1) \sim \log 2/\log n$ , and  $\sum (\log n)^{-1}$  diverges. This is compatible with the conjecture, since one bad prime contributes only  $O(\log n)$  to  $\log G_n$ . To violate  $G_n = e^{o(n)}$ , one needs  $B(n) \geq \varepsilon n/(6 \log n)$ ; the Poisson tail gives  $\mathbb{P}(B(n) \geq k_n) \leq e^{-cn}$  for  $k_n = \lceil \varepsilon n/(6 \log n) \rceil$ , and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

## 10 Computational verification

We computed  $G_n$  exactly for  $n = 1, \dots, 1000$  using rational arithmetic.

| $n$ range | avg $\log(G_n)/n$ | max $\log(G_n)/n$ |
|-----------|-------------------|-------------------|
| 1–50      | 0.276             | 0.93              |
| 51–100    | 0.108             | 0.26              |
| 101–200   | 0.086             | 0.18              |
| 201–300   | 0.048             | 0.10              |
| 301–400   | 0.038             | 0.08              |
| 401–500   | 0.024             | 0.05              |
| 501–600   | 0.029             | 0.06              |
| 601–700   | 0.017             | 0.04              |
| 701–800   | 0.014             | 0.04              |
| 801–900   | 0.016             | 0.04              |
| 901–1000  | 0.013             | 0.03              |

A power-law fit gives  $\log(G_n)/n \sim 5n^{-0.87}$ , i.e.,  $\log G_n \sim 5n^{0.13}$ , far stronger than the density-one bound  $o(n)$  of Theorem 42.

The exponent 0.13 should be interpreted with caution: if  $\log G_n \sim \kappa \log n$ , the effective log-log slope is  $1/\log n = 0.145$  at  $n = 1000$ , close to the fitted 0.13. A random-gcd model—assigning each prime  $p \leq n$  an independent probability  $\sim 1/p$  of dividing  $G_n$ —predicts  $\mathbb{E}[\log G_n] \sim \log n$  by the weighted Mertens theorem, with standard deviation  $\sim (\log n)/\sqrt{2}$  of the same order. The data thus cannot yet distinguish  $\kappa \log n$  from  $n^{0.13}$ ; the logarithmic prediction is arithmetically more natural.

## 11 Remarks

1. The Beukers supercongruence  $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$  [4] provides tower rigidity: for each fixed prime,  $v_p(b_n)$  is controlled by the base- $p$  digits. Quantitatively, since  $b_n \equiv \prod b_{d_i} \pmod{p}$  (Theorem 4), we have  $p \nmid b_n$  iff every base- $p$  digit of  $n$  avoids  $\mathcal{Z}_p$ . Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial  $C_2(m) = P(m+1)$ , the factorization  $P(x) = (2x+1)(17x^2+17x+5)$  shows that gap-2 pairs can only occur at  $m = (p-3)/2$  (when  $2(m+1)+1 = 2m+3 \equiv 0$ ) or at roots of the quadratic  $17x^2+17x+5$  (discriminant  $-51$ ; exists over  $\mathbb{F}_p$  iff  $(\frac{-51}{p}) = 1$ ). Empirically, all observed gap-2 pairs arise from the linear factor at  $m = (p-3)/2$ .
3. For  $1 \leq j \leq p-2$ , the Apéry number  $b_j \bmod p$  admits a character-sum representation as a Greene [12] Gaussian hypergeometric  ${}_4F_3$  over  $\mathbb{F}_p$  with parameter  $\chi = \omega^j$  (Teichmüller character). However, the Weil bound controls the archimedean size  $|b_j|_\infty = O(p^{3/2})$  but does *not* bound  $Z(p)$ , since zeros of  $b_j$  modulo  $p$  are a  $p$ -adic event. The gap polynomial argument above is specifically adapted to this  $p$ -adic zero-count problem.
4. Malik–Straub [15] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [18] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our  $Z(p)$  data for the  $\zeta(3)$  sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial  $H_p(t) = \sum_{j=0}^{p-1} b_j t^j \in \mathbb{F}_p[t]$  is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in  $\mathbb{F}_p$  are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic  $Z(p)$  counts vanishing *coefficients* of  $H_p$ , equivalently the vanishing finite Mellin coefficients  $b_j = -\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j}$  for  $1 \leq j \leq p-2$ . Thus  $Z(p)$  is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent  $F$ -crystal theory do not by themselves bound  $Z(p)$ ; they control the root locus, not the Fourier support. A bounded-complexity geometric lift *does* exist: the lisse sheaf  $\mathcal{V}_\ell = R^2 f_* \mathbb{Q}_\ell$  on the smooth locus of the K3 pencil (rank 22, conductor independent of  $p$ ) satisfies  $\text{Tr}(\text{Frob}_t \mid \mathcal{V}_\ell) \equiv A_p(t) \pmod{p}$  by the Hasse–Witt congruence. The Mellin sum  $\mathcal{M}_{p,j} = \sum_t \tau_p(t) \omega_p(t)^{-j}$  is then realized on  $R\Gamma_c(U_{\mathbb{F}_p}, \mathcal{V}_\ell \otimes \mathcal{L}_{\omega_p^{-j}})$  with bounded rank and conductor. However, the event  $b_j \equiv 0 \pmod{p}$  is the *same-characteristic* divisibility  $\mathfrak{p}_p \mid \mathcal{M}_{p,j}$ , a crystalline condition outside the scope of  $\ell$ -adic equidistribution (Forey–Fresán–Kowalski [11]). The missing theorem is a *crystalline Mellin anti-concentration* estimate controlling  $\#\{j : p \mid \mathcal{M}_{p,j}\}$ .
6. **Jacobi counterexample: bounded complexity is not enough.** Fix an integer  $d \geq 2$ . For every prime  $p \equiv 1 \pmod{d}$ , the rank-1 Kummer sheaf  $\mathcal{L}_\psi$  on  $\mathbb{P}^1 \setminus \{0, 1, \infty\}$  attached to  $\psi = \omega_p^{-(p-1)/d}$  (order  $d$ ) has Mellin transforms  $M_j = J(\omega_p^{-j}, \psi)$ . The Gross–Koblitz/Stickelberger valuation  $v_p(g(\omega^{-k})) = k/(p-1)$  and the Jacobi factorization  $J(\alpha, \beta) = g(\alpha)g(\beta)/g(\alpha\beta)$  give  $v_p(M_j) = 1$  whenever  $j \geq (p-1)(d-1)/d$ , yielding at least  $(p-1)/d - 1$  characters with  $p \mid M_j$ . Thus a sheaf of *fixed* rank, conductor, and Weil bound  $O(\sqrt{p})$  can have  $\gg p$  bad tame characters. This shows that bounded cohomological complexity alone cannot control  $Z(p)$ ; any proof of Hypothesis  $Z$  must use Apéry-specific arithmetic.
7. **Sufficient criteria for Hypothesis  $Z$ .** Two clean algebraic conditions, either of which would imply  $Z(p) = O(1)$ :
  - (a) *Bounded tame-Hasse polynomial.* Suppose there exist an absolute integer  $D$  and, for each sufficiently large  $p$ , a nonzero Laurent polynomial  $Q_p \in \mathbb{F}_p[X, X^{-1}]$  of degree at most  $D$  and units  $u_{p,j} \in \mathbb{F}_p^\times$  such that  $\mathcal{M}_{p,j} \equiv u_{p,j} Q_p(\omega_p^j) \pmod{p}$  for every nontrivial tame character. Then  $Z(p) \leq 2D$ , since  $Q_p$  (after clearing negative exponents) has at most  $2D$  roots.
  - (b) *Bounded nonordinary twists.* Suppose for each  $p$  the set  $E_p$  of tame characters  $\chi$  for which the twisted crystal  $H_{\text{rig},c}^1(U, \mathcal{V} \otimes \mathcal{K}_\chi)$  fails to have a unique unit-root eigenvalue satisfies  $|E_p| \leq C$ . Then  $Z(p) \leq C + O(1)$ , since a unique unit root forces  $v_p(\mathcal{M}_{p,j}) = 0$ .
 Criterion (a) is the algebraic version of Hypothesis  $Z$ ; criterion (b) is its  $p$ -adic-slopes formulation. The Jacobi counterexample above shows that neither criterion holds for general bounded-complexity sheaves.
8. **Palindromic symmetry.** For  $0 \leq j \leq p-1$  and  $0 \leq k \leq p-1-j$ , the identity  $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod{p}$  holds (each of the  $k$  falling factors  $p-1-j-i \equiv -(j+1+i)$ ). When  $k \leq j$ ,  $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$ ; when  $k > j$ , Lucas’s theorem gives  $\binom{p-1-j+k}{k} \equiv 0$  (since  $p-1-j+k \geq p$  and  $\binom{k-j-1}{k} = 0$ ). Therefore each term of  $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$  with  $k > j$  vanishes, and for  $k \leq j$  the  $(-1)^{2k}$  signs cancel, giving  $b_{p-1-j} \equiv b_j \pmod{p}$ . The extra terms of  $b_j$  when  $j > (p-1)/2$  (i.e.,  $k > p-1-j$ ) likewise vanish:  $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$  since  $j+k-p < k$ . Equivalently,  $H_p(t)$  is palindromic:  $t^{p-1} H_p(1/t) \equiv H_p(t)$ . This explains the observed pairing: whenever  $b_j \equiv 0$ , also  $b_{p-1-j} \equiv 0$ , so zeros of  $b_j$  come in pairs  $\{j, p-1-j\}$  (except at the self-paired center  $j = (p-1)/2$ , which vanishes only at non-ordinary primes).

Proposition 50 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo  $p$ , in particular the companion solution  $a$ . The “pair count”  $K(p) = Z(p)/2$  (or  $(Z(p) - 1)/2$  at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 28).

9. **Sym<sup>2</sup> squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [20]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [7] proved the mod- $p$  squareness rigorously: for every prime  $p \geq 5$ , setting  $\varepsilon_p = \frac{1}{2}(1 - (\frac{-6}{p}))$ , there exists a unique (up to sign) polynomial  $B_p \in \mathbb{F}_p[t]$  such that

$$H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where  $\Delta(t) = t^2 - 34t + 1$  is the conifold discriminant and  $B_p$  is squarefree and coprime to  $\Delta$ . The proof uses the explicit identity  $F(t(x)) = (1+x)h(x)^2$  between the Apéry and Franel generating functions, the  $p$ -Lucas property of both sequences, and a quadratic-descent argument through the covering  $\mathbb{F}_p(x)/\mathbb{F}_p(t)$  where  $t = x(1-8x)/(1+x)$ . The deck-transformation character  $(-6/p)$  determines whether  $\varepsilon_p = 0$  or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \quad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently,  $\varepsilon_p = 0$  iff  $p \equiv 1, 5, 7, 11 \pmod{24}$ . These two cases occur with equal density 1/2 by Dirichlet’s theorem.

Splitting the mean  $Z(p)$  by residue class modulo 8 (and modulo 24) yields no significant difference (mean  $\approx 1.00$  in each class,  $n \geq 2000$  primes per class up to  $10^5$ ); the squareness structure does not directly influence  $Z(p)$ , consistent with  $Z(p)$  counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure:  $b_j$  is (up to the  $\Delta^{\varepsilon_p}$  correction) the  $j$ -th coefficient of  $B_p^2$ , i.e.,  $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$  where  $\beta_k$  denotes the coefficients of  $B_p$ . The vanishing  $b_j \equiv 0$  thus requires cancellation across  $\sim p/2$  convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree  $(p-1)/2$  over  $\mathbb{F}_p$ , for  $p \in \{101, 503, 1009, 5003\}$ ) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean  $\approx 1$ , variance  $\approx 1$ , total variation distance to Poisson(1) equal to 0.018). However, the observed  $Z(p)$  distribution over 667 primes  $p \leq 5000$  is *not* Poisson(1) (TV = 0.43); instead it matches  $2 \cdot \text{Poisson}(1/2)$  with TV = 0.023. The explanation is the palindromic pairing: since  $b_{p-1-j} \equiv b_j$ , zeros come in pairs  $\{j, p-1-j\}$  and  $Z(p)$  is almost always even. Denoting by  $Z_{\text{pair}}(p)$  the number of zero pairs, the data show  $Z_{\text{pair}} \sim \text{Poisson}(1/2)$ , so  $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$ . Thus squareness explains the  $O(1)$  mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts  $\max_{p \leq N} Z(p) \sim 2 \log \log N$  via extreme-value theory, strongly suggesting that  $Z(p)$  is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of  $B_p$  imposes a bounded *average* (Hypothesis  $\bar{Z}$ ) remains the central open question.

The factorization does *not* improve the divisor-counting bound  $D(P, Q) \leq Q^2/\log P$ . In the regime  $P \geq Q^{4/3}$  where the divisor estimate is active, one has  $q < p/2$ , so the coefficient  $b_q \pmod{p}$  lies in the formally free triangular range of the square root: the map  $(\beta_1, \dots, \beta_q) \mapsto (b_1, \dots, b_q) \pmod{p}$  is a triangular automorphism with Jacobian  $2^q$ , hence the single equation  $b_q \equiv 0$  imposes no algebraic constraint beyond fixing one coefficient of  $B_p$ . Thus the Sym<sup>2</sup> structure gives the correct *random model* (uniform square root  $\Rightarrow$  Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

10. **Toric framework.** The Laurent polynomial  $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$  on  $(\mathbb{F}_p^\times)^4$  has Newton polytope  $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$ , a 4-dimensional pyramid over the regular octahedron in  $\mathbb{R}^3$ . Its  $f$ -vector is  $(7, 18, 20, 9)$ , its normalized volume is  $4! \cdot \text{vol}(\Delta) = 8$ , and it is Kouchnirenko nondegenerate for every prime  $p \geq 5$ : for each of the 27 nonempty faces  $\sigma$  not containing the origin, the toric critical system  $\{x_i \partial f_\sigma / \partial x_i = 0\}$  has no solution in  $(\mathbb{F}_p^\times)^4$ . For the Adolphson–Sperber twisted Hodge polygon with  $w$ -Kummer twist  $\chi_w = \omega^{-j}$  (character parameter  $a = j/(p-1)$ ), the slopes are  $a, a+1, a+2, a+3$  with multiplicities  $1, 3, 3, 1$ , confirming that the first slope equals  $a$  for every  $a \in \{k/100 : 0 \leq k \leq 99\}$ . By Newton-above-Hodge (Adolphson–Sperber [2]), the  $p$ -adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The  $Z(p)$  bound thus reduces to proving ordinarity for all but  $O(1)$  twist parameters  $j$ .
11. **Branch obstruction.** The  $p-1$  Kummer twists  $\chi^{-j}$  ( $j = 0, \dots, p-2$ ) are distinct characters of conductor  $p$ . In Iwasawa-theoretic coordinates  $\chi^j = \omega^a \langle \cdot \rangle^s$  with  $a = j \bmod (p-1)$ , the horizontal family  $\{j : 0 \leq j < p-1\}$  hits *each* of the  $p-1$  branches  $\omega^a$  exactly once. Weierstrass preparation and bounded- $\lambda$  rigidity control zeros *within one branch* (the vertical direction  $j, j + (p-1), j + (p-1)p, \dots$ , where Dwork congruences live) but not *across* branches. Thus there is no  $p$ -adic analytic function of one variable whose zero count majorizes  $Z(p)$ . Any “rigidity” proof of Hypothesis  $\bar{Z}$  would need a mechanism not supplied by existing Iwasawa–Dwork theory.
12. Any estimate  $\log G_n = o(n)$ —whether  $O(\log n)$ ,  $O(\sqrt{n})$ , or  $O(n^{0.13})$ —yields  $\gamma = 0$  in the exponent  $G_n = e^{\gamma n + o(n)}$  and therefore does not improve the Rhin–Viola irrationality measure  $\mu(\zeta(3)) \leq 5.513$  [19], which depends only on the exponential growth rate  $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$ . An improvement would require an exponentially large common divisor, which is not predicted by any current model.
13. **Comparison with previous work.** Classical refinements of Apéry’s method (Beukers [4], Rhin–Viola [19], Zudilin [21]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for every index. They do not provide exceptional-set estimates for the *actual* reduced gcd  $G_n$ . A separate body of work, based on the Subspace Theorem, proves  $\gcd(a^n - 1, b^n - 1) = e^{o(n)}$  pointwise for multiplicatively independent  $a, b$  (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence’s polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.
14. **The remaining open problem.** The companion-block height bound (Proposition 51) shows that primes  $p > \sqrt{n}$  with  $D_q \equiv 0 \pmod{p}$  contribute  $O(n^{2/3})$  to  $\log G_n$  at *every* index. After removing this channel, the conjecture  $G_n = e^{o(n)}$  for all  $n$  is equivalent to the *pointwise radical estimate*

$$\sum_{\substack{p > \sqrt{n} \\ p | b_n \bmod p}} \log p = o(n).$$

Theorem 53 proves this for all but  $O_\varepsilon((\log N)^2)$  integers in  $[1, N]$ . However, the codegree method (Lemma 52) controls primes *shared* between pairs of integers, and does not bound the total weight of “private” primes at a single index. Closing the gap requires a one-point

rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes.

## 12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving  $W(n) = o(n)$  for *every*  $n$ —we study the *second moment* of the bad-prime weight function.

### 12.1 The criterion

For a dyadic shell  $I_N = (N, 2N]$ , define

$$B(m) = \sum_{\substack{p > \sqrt{m} \\ p \mid b_m \bmod p}} \log p, \quad M_2(N) = \sum_{m \in I_N} B(m)^2.$$

**Proposition 61** (Second-moment criterion). *If  $M_2(N) \leq C \cdot N^{2-\delta}$  for some  $C > 0$  and  $\delta > 0$ , then  $W(n) = o(n)$  for all  $n$ . More precisely, for every  $n \in I_N$ ,*

$$W(n) \leq B(n) \leq \sqrt{M_2(N)} \leq C^{1/2} \cdot N^{1-\delta/2}.$$

*Proof.*  $B(n)^2$  is one nonneg. term of the sum  $M_2(N)$ , hence  $B(n) \leq \sqrt{M_2(N)}$ .  $\square$

In particular,  $M_2(N) = O(N \cdot (\log N)^A)$  for any fixed  $A$  gives  $B(n) = O(\sqrt{N} \cdot (\log N)^{A/2}) = o(N)$ .

*Remark 62* (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of  $p \mapsto n \bmod p$  on primes  $p > \sqrt{n}$  (two such primes giving the same residue would force their product  $\leq n$ ),

$$W(n) = \sum_{0 < r < n} \log \text{rad}_{>\max(\sqrt{n}, r)} \gcd(n-r, b_r).$$

Each inner radical is 1 or a single prime  $p = n-r$ . For the top-half block  $p > n/2$ , the condition becomes “ $n-r$  is prime and  $(n-r) \mid b_r$ .”

**Proposition 63** (Hard-core reduction). *For each bad prime  $p > \sqrt{n}$ , write  $n = q_p p + r_p$ ,  $s_p = \min(r_p, p-1-r_p)$ ,  $h_p = |p-1-2r_p|$ . Let  $1 \leq Q, R, H < n/2$ . Then the total weight of primes satisfying at least one of  $q_p > Q$ ,  $s_p \leq R$ ,  $h_p \leq H$  is  $O(n/Q + (R+H) \log n)$ . In particular, with  $Q = R = H = \sqrt{n/\log n}$ , all “easy channels” contribute  $O(\sqrt{n \log n})$ ; the remaining hard core consists of primes  $p \gg \sqrt{n \log n}$  with zero positions far from both endpoints and the reflection centre.*

*Proof.* Large quotient ( $q_p > Q$ ): then  $p < n/Q$ , so  $\sum \log p \leq \vartheta(n/Q) \ll n/Q$ . Near endpoint ( $s_p \leq R$ ): by the injectivity of  $p \mapsto n \bmod p$ , at most two primes  $> \sqrt{n}$  contribute for each value of  $s_p$ . Near centre ( $h_p \leq H$ ): the identity  $(2q_p+1)p = 2n+1 \pm h_p$  forces  $p$  to divide a fixed integer of size  $O(n)$ , giving  $O(1)$  primes per value of  $h_p$ .  $\square$

*Remark 64* (The vertical main term is harmless). Write  $W(n) = M(n) + \mathcal{E}(n)$  where  $M(n) = \sum_{p > \sqrt{n}} Z(p) (\log p)/p$ . The unconditional bound  $Z(p) \leq Cp^{2/3}$  gives  $M(n) \ll \sum_{p \leq n} p^{-1/3} \log p \ll n^{2/3}$ . The entire problem is the oscillatory error  $\mathcal{E}(n)$ , which at each prime is  $(\log p) \cdot (1_{n \bmod p \in \mathcal{Z}_p} - Z(p)/p)$ .

## 12.2 Empirical verification

The following table, computed for all dyadic shells with  $64 \leq N \leq 4096$  using the full zero-set data for all primes  $p \leq 20,000$ , gives:

| $N$  | $M_2(N)$ | $M_2/N$ | $M_2/(N \log^2 N)$ | $\max B(m)$ |
|------|----------|---------|--------------------|-------------|
| 64   | 672      | 10.5    | 0.605              | 10.4        |
| 128  | 1,684    | 13.2    | 0.560              | 15.9        |
| 256  | 4,376    | 17.1    | 0.555              | 15.8        |
| 512  | 11,896   | 23.2    | 0.598              | 22.3        |
| 1024 | 28,632   | 28.0    | 0.583              | 21.7        |
| 2048 | 69,226   | 33.8    | 0.582              | 27.1        |
| 4096 | 166,671  | 40.7    | 0.587              | 32.1        |
| 8192 | 421,868  | 51.5    | 0.634              | 37.2        |

The ratio  $M_2(N)/(N \log^2 N)$  stabilizes near 0.59, strongly suggesting

$$M_2(N) \sim C_2 \cdot N \cdot (\log N)^2, \quad C_2 \approx 0.59. \quad (15)$$

This gives  $B(n) \leq \sqrt{0.59 N} \log N \approx 0.77 \sqrt{N} \log N = o(N)$ .

## 12.3 CRT expansion

Expand  $M_2$  dyadically. For distinct primes  $p, q \in (P, 2P]$  with  $P > \sqrt{N}$ , write  $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$ . Then

$$M_2(N) = \underbrace{\sum_p (\log p)^2 |\Omega_p|}_{\text{diagonal}} + \underbrace{\sum_{p \neq q} \log p \log q |\Omega_p \cap \Omega_q|}_{\text{off-diagonal}}.$$

**Diagonal.**  $|\Omega_p| = Z(p) \cdot N/p + O(Z(p))$ , so

$$\text{diag} = N \sum_p \frac{Z(p) (\log p)^2}{p} + O\left(\sum_p Z(p) (\log p)^2\right).$$

Under the Poisson model  $\mathbb{E}[Z(p)] = 1$ , the main term is  $\sim N \cdot \sum_{P < p \leq 2P} (\log p)^2/p \sim N \cdot (\log P)^2 \cdot \log 2$ . Summing dyadically  $P = \sqrt{N}, 2\sqrt{N}, \dots, N$  gives diagonal  $\sim \frac{1}{2} N (\log N)^2$ , matching  $\approx 63\%$  of the empirical (15).

**Off-diagonal.** Since  $p, q > \sqrt{N}$  we have  $pq > N$ , so each CRT class  $r_p \pmod{p}$ ,  $r_q \pmod{q}$  contains at most one element of  $I_N$ . The CRT main term of the off-diagonal is  $\sim N \left(\sum Z(p) \log p/p\right)^2 \sim \frac{1}{4} N (\log N)^2$ .

**The predicted constant.** Under the independent random model—each prime  $p$  hits a uniform  $m$  with probability  $\sim 1/p$ , independently across primes—the second moment of  $B(m)$  at one integer is  $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \text{Var}(B)$ . By weighted Mertens,  $\mathbb{E}[B] = \sum_{\sqrt{m} < p \leq m} (\log p)/p = \frac{1}{2} \log m + O(1)$ , and

$$\text{Var}(B) = \sum_{\sqrt{m} < p \leq m} \frac{(\log p)^2}{p} \left(1 - \frac{1}{p}\right) = \frac{3}{8} (\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] = \left(\frac{1}{4} + \frac{3}{8}\right)(\log m)^2 + O(\log m) = \frac{5}{8}(\log m)^2 + O(\log m), \quad (16)$$

predicting  $M_2(N) \sim \frac{5}{8} N (\log N)^2$ . The diagonal constant is  $3/8$  (matching the observed  $\text{Diag}/N \log^2 N \approx 0.37$ ) and the off-diagonal contributes  $1/4$ . The empirical ratio  $M_2/(N \log^2 N) \approx 0.59$  is consistent with convergence to  $5/8 = 0.625$ , with lower-order boundary effects.

**Centered variance and cross-prime covariance.** The centered dispersion  $V^\circ = M_2 - M_1^2/N$  decomposes as

$$V^\circ = \underbrace{\sum_p (\log p)^2 |\Omega_p| (1 - |\Omega_p|/N)}_{\leq \text{Diag}} + \underbrace{\sum_{p \neq q} \log p \log q (|\Omega_p \cap \Omega_q| - |\Omega_p| |\Omega_q|/N)}_{E_N^\circ}. \quad (17)$$

The first sum is at most the diagonal  $S$  (since  $|\Omega_p|/N \geq 0$ ). The term  $E_N^\circ$  is the centered cross-prime covariance: it vanishes exactly when the events  $\{m \in \Omega_p\}$  are pairwise uncorrelated. The computed values are:

| $N$   | $M_2/N \log^2 N$ | $\text{Diag}/N \log^2 N$ | $V^\circ/S$ | $E_N^\circ$ |
|-------|------------------|--------------------------|-------------|-------------|
| 256   | 0.556            | 0.359                    | 1.014       | +110        |
| 512   | 0.597            | 0.376                    | 1.010       | +215        |
| 1024  | 0.582            | 0.375                    | 0.981       | -152        |
| 2048  | 0.581            | 0.370                    | 0.992       | +7          |
| 4096  | 0.588            | 0.371                    | 0.995       | +41         |
| 8192  | 0.634            | 0.387                    | 0.995       | -188        |
| 16384 | 0.669            | 0.400                    | 0.995       | -922        |
| 32768 | 0.646            | 0.394                    | 0.988       | -14065      |

The ratio  $V^\circ/S$  converges to 1 from below for  $N \geq 1024$ , with  $V^\circ/S < 1$  at every scale tested beyond 512. The negative sign of  $E_N^\circ$  for large  $N$  reflects genuine *anti-correlation* among the CRT-constrained incidence events: at  $N = 1024$ , over 96% of cross-prime covariance pairs  $C_{p,q}$  are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude  $|E_N^\circ|$  is  $O(\sqrt{N} (\log N)^2)$ , so  $|E_N^\circ|/S \rightarrow 0$  at rate  $\sim 1/\sqrt{N}$ .

**Three-model comparison.** To test whether the observed  $V^\circ/S \approx 1$  is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block  $I_N$ : (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of  $\mathbb{F}_p^*$  with the same cardinalities  $|Z_p|$ ; (3) random reflected subsets with cardinalities drawn from  $2 \cdot \text{Poisson}(1/2)$ . Over 20 independent trials of each random model and  $N \in \{256, 512, 1024, 2048\}$ , all three models yield  $V^\circ/S = 1.00 \pm 0.05$ , with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

*Remark 65* (Boundary correction). The naïve CRT main term  $Z(p)Z(q)N/(pq)$  overestimates the random prediction for primes near  $\sqrt{N}$  or  $N$ , because the admissible interval  $J_{p,q} = J_p \cap J_q$  may be shorter than  $N$ . After replacing  $N$  by  $L_{p,q} = |J_{p,q}|$  in the main term, the boundary-corrected CRT residual is positive and  $O(\sqrt{N} (\log N)^2)$ —the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

## 12.4 The no-go model

Define the *adversarial zero-sets*: for one fixed  $m_0 \in I_N$ , set  $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$ . Then  $|\mathcal{Z}_p^*| = 2$  (satisfying reflection), yet  $m_0 \in \Omega_p^*$  for *every* prime  $p > \sqrt{N}$ , giving  $B^*(m_0) = \sum_{p > \sqrt{N}} \log p = n - o(n)$ . In particular:

- The adversarial model satisfies  $Z^*(p) = 2$ ,  $Z_{\text{pair}}^*(p) = 1$ , and all proved vertical bounds.
- Yet  $M_2^*(N) \geq B^*(m_0)^2 \geq cn^2$ , which is  $\Omega(N^2)$ , not  $O(N)$ .

**Conclusion:** *No proof using only vertical information (the sizes  $Z(p)$ , the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish  $W(n) = o(n)$  for all  $n$ . The missing theorem must use cross-prime horizontal information: the locations of the zeros  $\mathcal{Z}_p$  across different primes  $p$ .*

## 12.5 The Fourier form of the CRT error

Define the modular Fourier transform of  $\mathcal{Z}_p$ :

$$S_p(a) = \frac{1}{p} \sum_{r \in \mathcal{Z}_p} e\left(\frac{ar}{p}\right).$$

Then  $S_p(0) = Z(p)/p$  and, by the palindromic pairing,  $e(a(p-1)/(2p)) S_p(a)$  is real-valued for every  $a$ . Fourier inversion gives

$$E_N(p, q) = \sum_{\substack{(a,b) \neq (0,0) \\ 0 \leq a < p, 0 \leq b < q}} S_p(a) S_q(b) \sum_{m \in I_N} e\left(\frac{am}{p} + \frac{bm}{q}\right).$$

The inner sum is a standard exponential sum  $|K_N(\alpha)| \leq \min(N, 1/\|\alpha\|)$  where  $\alpha = a/p + b/q$ .

Applying the classical additive large sieve to bound  $\sum_{p \neq q} |E_N(p, q)|$  gives  $O(N^2/\log N)$  at the top prime scale  $p \sim N$ —one full logarithm larger than the threshold  $o(N^2/\log^2 N)$  required for  $M_2 = O(N(\log N)^A)$ . Thus the classical large sieve does not close the gap.

## 12.6 The dispersion target

Define the centered dispersion

$$V^\circ(P, N) = \sum_{m \in I_N} \left| B_P(m) - \frac{S(P, N)}{N} \right|^2,$$

where  $B_P(m) = \sum_{P < p \leq 2P} 1_{\Omega_p}(m)$  and  $S(P, N) = \sum_p |\Omega_p|$ .

**Hypothesis 66** (AP–BDH dispersion).  $V^\circ(P, N) \ll N^{o(1)} \cdot S(P, N)$ .

**Proposition 67.** *Hypothesis 66 together with the unconditional bound  $Z(p) = O(p^{2/3})$  implies  $W(n) = o(n)$  for all  $n$ .*

*Proof.* By hypothesis, for every  $m \in I_N$ ,

$$B_P(m) \leq \frac{S(P, N)}{N} + \sqrt{V^\circ(P, N)} \leq \frac{S(P, N)}{N} + N^{o(1)} \sqrt{S(P, N)}.$$

By the zero-count bound,  $S(P, N) \leq (N/P + 1) \cdot \sum_{P < p \leq 2P} Z(p) \ll NP^{2/3}/\log P$ . Therefore  $B_P(m) \ll N^{o(1)} \cdot (NP^{2/3}/\log P)^{1/2}$ . The weighted sum satisfies  $W_P(m) \leq B_P(m) \cdot \log(2P)$ . Summing over  $O(\log N)$  dyadic blocks gives

$$W(n) \ll N^{o(1)} \cdot N^{1/2} \cdot N^{1/3} = N^{5/6+o(1)} = o(N). \quad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets  $\mathcal{Z}_p$  do not systematically align at one index across different primes. The no-go model of Section 12.4 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

*Remark 68* (Covariance sign distribution). At  $N = 1024$ , more than 93% of the  $\binom{113}{2} = 6,328$  cross-prime covariance terms  $C_{p,q}$  are negative. This negativity has a clean structural explanation: for primes  $p, q > \sqrt{N}$ , the Poisson parameter  $\lambda_{p,q} = A_p A_q / N < 1$ , so  $C_{p,q} < 0$  precisely when the joint count  $J_{p,q}$  vanishes. In the sparse regime,  $J_{p,q} = 0$  for  $e^{-\lambda} \approx 96\%$  of pairs, while the rare pairs with  $J_{p,q} = 1$  have  $C_{p,q} = 1 - \lambda_{p,q} \approx 1$ . Over a complete period  $(\bmod pq)$ , the average of  $C_{p,q}$  is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal  $E^\circ = \sum_{p \neq q} w_p w_q C_{p,q}$  oscillates in sign: positive at  $N = 1024, 4096, 8192$  and negative at  $N = 2048$  (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate  $|E^\circ| = o(S)$ , which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson’s inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

*Remark 69* (The dispersion route bypasses  $\bar{Z}$ ). Hypothesis 66 does *not* require Hypothesis  $\bar{Z}$ . The adversarial singleton model  $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$  satisfies  $\bar{Z}$  in its strongest form ( $Z(p) = 2$  for all  $p$ ), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with  $Z(p) = O(p^{2/3})$  already gives  $W(n) = o(n)$ , even without any average bound on  $\sum Z(p)$ . This makes the dispersion route *strictly stronger* than the  $\bar{Z}$  route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that  $V^\circ/S \approx 1$  empirically, matching the prediction of the independent random model.

*Remark 70* (Martingale decomposition). Order the active primes as  $p_1 < \dots < p_K$  and define the Doob martingale  $M_k = \mathbb{E}[B \mid X_{p_1}, \dots, X_{p_k}]$ . The increments  $D_k = M_k - M_{k-1}$  are orthogonal:  $\text{Var}(B) = \sum_k \mathbb{E}[D_k^2]$ . However,  $D_k = (X_{p_k} - \pi_k) \Delta_k$  where the “predictive cascade”  $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j} (\mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=1) - \mathbb{P}(X_{p_j}=1 \mid \mathcal{F}_{k-1}, X_{p_k}=0))$  contains all future primes. Under independence  $\Delta_k = w_{p_k}$  and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that  $m \bmod p_k$  falls in a zero can nearly identify  $m$  (when  $A_{p_k} \approx 1$ ), making most subsequent indicators nearly deterministic. Controlling the cascade  $\sum \mathbb{E}[\pi_k(1 - \pi_k)\Delta_k^2] \leq (1 + o(1)) \sum w_{p_k}^2 \alpha_{p_k}$  is equivalent to the dispersion inequality itself.

*Remark 71* (Concentration and the all- $n$  gap). The  $L^2$ -to-uniform upgrade (Lemma 73) gives  $\max |B(m) - \mu| \leq \sqrt{V^\circ}$ . If  $V^\circ \leq C S$  for some constant  $C$ , then  $\sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$ , so the all- $n$  theorem  $W(n) = o(n)$  follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 74). Empirically, the actual concentration is much stronger:  $\max B(m)/(N/\log N) \approx 0.07$  at  $N = 4096$  (with  $\max B \approx 36$  vs.  $N/\log N \approx 492$ ), and the maximum  $|B(m) - \mu|/\sigma$  stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

*Remark 72* (Block decomposition of  $V^\circ$ ). Decompose the prime sum  $B(m) = \sum_j B_{P_j}(m)$  by dyadic prime blocks  $P_j < p \leq 2P_j$ . Then  $V^\circ = \sum_j V_{P_j}^\circ + 2 \sum_{j<k} \text{cross}(P_j, P_k)$ . At  $N = 4096$ , the cross-block covariance is 0.03% of the full  $V^\circ$ . This is because cross-block pairs  $(p, q)$  with  $p \in (P, 2P]$ ,  $q \in (Q, 2Q]$ ,  $Q \geq 2P$  satisfy  $pq \geq 2P^2 > 2N$ , so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to  $V_P^\circ \ll N^{o(1)} S_P$  for each dyadic block *separately*—a within-scale CRT independence statement.

## 12.7 The covariance structure and the $L^2$ -to-uniform upgrade

| $N$  | $K$ | $V^\circ/S$ | $E^\circ$ | $ E^\circ /S$ | $J=0$ | $J=1$ |
|------|-----|-------------|-----------|---------------|-------|-------|
| 512  | 67  | 0.998       | +130.6    | 0.015         | 95.6% | 4.1%  |
| 1024 | 113 | 0.998       | +215.6    | 0.010         | 96.6% | 2.9%  |
| 2048 | 205 | 0.994       | +96.2     | 0.002         | 98.1% | 1.8%  |
| 4096 | 380 | 0.996       | +120.0    | 0.001         | 98.9% | 1.1%  |
| 8192 | 742 | 1.001       | +1344     | 0.005         | 99.3% | 0.6%  |

Table 1: Covariance structure of the off-diagonal sum  $E^\circ = V^\circ - S_c$ , computed over all  $\binom{K}{2}$  pairs of active primes  $p > \sqrt{N}$ .  $J=0$  and  $J=1$  are the fractions of pairs  $(p, q)$  with joint occupancy  $J_{p,q} = 0$  or 1. The Cauchy–Schwarz bound on  $|E^\circ|$  exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

**Lemma 73** ( $L^2$ -to-uniform upgrade). *Let  $F_N: I_N \rightarrow \mathbb{R}$  with  $\mu_N = \frac{1}{N} \sum_{m \in I_N} F_N(m)$  and  $\mathcal{V}_N = \sum_{m \in I_N} (F_N(m) - \mu_N)^2$ . Then*

$$\max_{m \in I_N} |F_N(m) - \mu_N| \leq \sqrt{\mathcal{V}_N}. \quad (18)$$

*Proof.* Each squared deviation  $(F_N(m_0) - \mu_N)^2$  is a single summand of  $\mathcal{V}_N$ , hence at most  $\mathcal{V}_N$ .  $\square$

**Corollary 74** (Dispersion implies the all-index Apéry bound). *If there exists a constant  $C$  such that  $V_w^\circ(N) \leq C S_w(N)$  for all sufficiently large dyadic  $N$ , where*

$$V_w^\circ(N) = \sum_{m \in I_N} (B(m) - \mu)^2, \quad S_w(N) = \sum_{p > \sqrt{N}} (\log p)^2 A_p,$$

*then  $W(n) = o(n)$  for all  $n$ , and consequently  $G_n = e^{o(n)}$  for every positive integer  $n$ .*

*Proof.* By Lemma 73 with  $F_N = B$ ,

$$\max_{m \in I_N} |B(m) - \mu| \leq \sqrt{V_w^\circ} \leq \sqrt{C S_w}.$$

Since  $S_w \leq (\log 2N)^2 \sum A_p$  and  $\sum A_p = O(N \log N)$  (from the bound  $Z(p) \leq Cp^{2/3}$  or even  $\sum Z(p) \ll \pi(x)$ ), we have  $S_w = O(N(\log N)^3)$ . Thus  $\sqrt{V_w^\circ} = O(\sqrt{N}(\log N)^{3/2})$ . The mean satisfies  $\mu = O((\log N)^2)$ . Therefore

$$W(n) \leq B(n) \leq \mu + \sqrt{V_w^\circ} = O(\sqrt{N}(\log N)^{3/2}) = o(N/\log N),$$

which gives  $W(n) = o(n)$  upon taking  $N = n$ .  $\square$

Table 1 and the extended computation of  $V^\circ/S$  through  $N = 32,768$  (Section 10) provide strong evidence for the bound  $V^\circ \leq C \cdot S$  with  $C \leq 1.01$ . The ratio  $|E^\circ|/S$  decreases from 0.015 at  $N = 512$  to 0.001 at  $N = 4096$ ; the brief increase at  $N = 8192$  is consistent with fluctuations of  $O(1/\sqrt{N})$ . No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the  $L^2$ -to- $L^\infty$  trivial inequality together with  $V^\circ = O(S)$  completes the proof.

## 12.8 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime  $p$ , retain the observed column size  $A_p$  and choose the incidence set  $\Omega_p$  uniformly among all  $A_p$ -subsets of  $I_N$ , independently across primes. Put  $D_p = A_p(1 - A_p/N)$  and  $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$ .

**Proposition 75** (Random CRT second moment). *Under the independent fixed-margin model,  $\mathbf{E} C_{p,q} = 0$  and  $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$ . Moreover,  $\mathbf{E}(C_{p,q} C_{p,r}) = 0$  for all triples of distinct primes. Consequently,*

$$\mathbf{E}[(E^\circ)^2] = \frac{2}{N-1} \left( D_w^2 - \sum_p w_p^4 D_p^2 \right). \quad (19)$$

If no single prime column carries a positive proportion of  $D_w$ , then

$$\frac{\text{sd}(E^\circ)}{S} \sim \sqrt{\frac{2}{N}}, \quad (20)$$

matching the empirical ratio  $|E^\circ|/S$  in Table 1 to within the expected fluctuations.

*Proof.* Given  $\Omega_p$ , the intersection count  $J_{p,q}$  is hypergeometric with parameters  $(N, A_p, A_q)$ , giving  $\mathbf{E} C_{p,q} = 0$  and  $\text{Var}(C_{p,q}) = D_p D_q / (N - 1)$ . For triples, conditioning on  $\Omega_p$  makes  $C_{p,q}$  and  $C_{p,r}$  conditionally independent with zero mean, so  $\mathbf{E}(C_{p,q} C_{p,r}) = 0$ . Expanding  $(E^\circ)^2$  and using orthogonality gives (19).  $\square$

## 12.9 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix  $H$  with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \neq q), \quad H_{p,p} = 0.$$

Then  $V^\circ = S_{\text{diag}} + v^T H v$  where  $v_p = w_p \sqrt{D_p}$  and  $|v|^2 = D_w$ . Consequently,  $V^\circ \leq (1 + \|H\|_{\text{op}}) S_{\text{diag}}$ .

| $N$  | $K$ | $\ H\ _{\text{op}}$ | $\lambda_{\text{max}}$ | $\lambda_{\text{min}}$ | Rayleigh( $v$ ) |
|------|-----|---------------------|------------------------|------------------------|-----------------|
| 256  | 39  | 0.824               | +0.824                 | −0.706                 | 0.022           |
| 512  | 67  | 0.987               | +0.987                 | −0.576                 | 0.031           |
| 1024 | 113 | 1.088               | +1.088                 | −0.707                 | 0.021           |
| 2048 | 205 | 0.896               | +0.896                 | −0.735                 | 0.004           |
| 4096 | 380 | 0.904               | +0.904                 | −0.734                 | 0.002           |
| 8192 | 742 | 1.210               | +1.210                 | −1.000                 | 0.009           |

Table 2: Spectral data for the normalized covariance matrix  $H$ . The operator norm  $\|H\|_{\text{op}}$  appears bounded (fluctuating between 0.8 and 1.2, not growing with  $N$ ). The Rayleigh quotient  $v^T H v / |v|^2 = E^\circ / D_w$  for the logarithmic weight vector  $v$  is much smaller and non-monotone with mean  $\approx 0.01$ , explaining why  $V^\circ \approx S$ . At  $N = 8192$ , two eigenvalues equal  $-1$  exactly, arising from a three-prime collision cluster with  $A_p \in \{1, 2, 3\}$ .

The striking gap between  $\|H\|_{\text{op}} \approx 1$  and the Rayleigh quotient  $\approx 0.01$  is explained by *eigenvector localization*. At every computed  $N$ , the top eigenvector concentrates on two or three primes with very sparse columns ( $A_p \leq 4$ ): these are large primes that share a common marked integer by

CRT coincidence. At  $N = 8192$  ( $\lambda_{\max} = 1.21$ ), the cluster consists of  $p = 14851, 7937, 7559$  with  $A_p = 1, 2, 3$  respectively. The logarithmic weight vector  $v_p = (\log p) \sqrt{D_p}$  assigns negligible mass to these sparse primes (combined contribution  $< 0.5\%$  of  $|v|^2$  at every  $N$ ), so it is nearly orthogonal to the localized extreme eigenvectors. At  $N = 4096$ , only  $3\%$  of  $|v|^2$  lies in the top 5 eigenvectors; the largest projection of  $v$  ( $10\%$ ) falls on an eigenvalue of magnitude  $0.14$ , deep in the spectral bulk. At  $N = 8192$ , the projection drops to  $0.5\%$ .

This localization has a combinatorial origin: the extreme eigenvalues of  $H$  arise from  $k$ -fold CRT collisions—integers  $m$  marked by  $k \geq 3$  distinct large primes. The corresponding eigenvectors span a  $(k-1)$ -dimensional space of eigenvalue  $-1$  and one direction of eigenvalue  $k-1$ . The logarithmic weight vector, being smoothly spread across all  $K$  active primes, is structurally immune to these localized collision artifacts.

## 12.10 Palm decorrelation and the cancellation mechanism

The identity (17) decomposes  $V^\circ = S_{\text{diag}} + E^\circ$  where the off-diagonal  $E^\circ = v^T H v$ . The Rayleigh quotient  $R_v = v^T H v / |v|^2 = E^\circ / D_w$  admits a transparent conditional-expectation interpretation. Define the *load excess*

$$\eta_p = \frac{(Hv)_p}{v_p} = \frac{\frac{1}{A_p} \sum_{m \in \Omega_p} (B_{-p}(m) - \mu_{-p})}{w_p (1 - A_p/N)}, \quad (21)$$

where  $B_{-p}(m) = B(m) - w_p X_p(m)$  is the total load excluding prime  $p$ , and  $\mu_{-p} = \mu - w_p A_p / N$ . The quantity  $\eta_p$  measures the average excess load from all *other* primes, conditional on  $m \in \Omega_p$ , normalized by the centered self-weight. Under exact independence,  $\eta_p = 0$  for every  $p$ . The Rayleigh quotient is the weighted average

$$R_v = \sum_p \frac{v_p^2}{D_w} \eta_p. \quad (22)$$

Thus  $V^\circ \approx S$  says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

*Remark 76* (Cancellation, not near-null). The small Rayleigh quotient does *not* mean that  $v$  is a near-null vector of  $H$ . Computation gives  $\|Hv\|/|v| \approx 0.31$  at all tested sizes  $N = 512, \dots, 4096$ , while  $R_v = v^T H v / |v|^2$  decreases from  $0.031$  to  $0.002$ . The cancellation indicator  $\|Hv\|/(|v| |R_v|)$  grows from  $12$  to  $154$ : positive and negative spectral contributions to  $R_v$  individually exceed  $0.13$  but nearly cancel. This is the standard trace-zero mechanism: since  $\text{tr } H = \sum_j \lambda_j = 0$  and the projections  $|\langle u, e_j \rangle|^2$  are approximately uniform ( $u = v/|v|$  is delocalized), the identity  $R_v = \sum_j \lambda_j (|\langle u, e_j \rangle|^2 - 1/K)$  produces cancellation to order  $K^{-1/2}$ . The extreme eigenvalues are localized (IPR of top eigenvector  $\approx 0.25$ , effective dimension  $\approx 4$ ), so their projections deviate from  $1/K$  but contribute little to the weighted sum.

## 12.11 Dyadic block compression

Partition the active primes into  $L = O(\log N)$  dyadic blocks  $\mathcal{B}_1, \dots, \mathcal{B}_L$  according to  $\lfloor \log_2 p \rfloor$ . For each block, define the restricted weight  $s_j = \|v|_{\mathcal{B}_j}\|$  and the unit share  $a_j = s_j / |v|$ . The *compressed covariance matrix*  $\tilde{H} \in \mathbb{R}^{L \times L}$  is

$$\tilde{H}_{i,j} = \frac{(v^{(i)})^T H v^{(j)}}{s_i s_j}, \quad (23)$$

where  $v^{(j)}$  is  $v$  restricted to  $\mathcal{B}_j$ . Then the Rayleigh quotient factors exactly as

$$R_v = a^T \tilde{H} a. \quad (24)$$

This reduces the  $K \times K$  problem to an  $L \times L$  one ( $L \leq 8$ ) in the single direction of interest.

| $N$  | $K$ | $L$ | $\ H\ _{\text{op}}$ | $\ \tilde{H}\ _{\text{op}}$ | $R_v$ | $\ Hv\ / v $ |
|------|-----|-----|---------------------|-----------------------------|-------|--------------|
| 512  | 67  | 6   | 0.987               | 0.113                       | 0.031 | 0.386        |
| 1024 | 113 | 6   | 1.088               | 0.111                       | 0.021 | 0.302        |
| 2048 | 205 | 7   | 0.896               | 0.097                       | 0.004 | 0.311        |
| 4096 | 380 | 7   | 0.904               | 0.040                       | 0.002 | 0.313        |
| 8192 | 742 | 8   | 1.210               | 0.052                       | 0.009 | 0.291        |

Table 3: Full operator norm  $\|H\|_{\text{op}}$  versus compressed operator norm  $\|\tilde{H}\|_{\text{op}}$ . The full norm is bounded near 1 with a spike to 1.21 at  $N = 8192$  (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at  $N = 512$  to 0.040–0.052 at  $N = 4096$ –8192. The column  $\|Hv\|/|v|$  shows that  $v$  is not a near-null vector of  $H$  ( $\|Hv\|/|v| \approx 0.3$  at all sizes); the small Rayleigh quotient arises from spectral cancellation.

The compressed norm  $\|\tilde{H}\|_{\text{op}}$  controls the Rayleigh quotient via  $|R_v| \leq \|\tilde{H}\|_{\text{op}}$ , and is both numerically accessible (an  $L \times L$  eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for  $K$  independent isotropic columns in  $\mathbb{R}^N$  predicts  $\|H\|_{\text{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$ ; with  $K \sim N/\log N$ , this gives  $2/\sqrt{\log N} + 1/\log N$ , which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low- $A_p$  columns). For the compressed matrix, the same model predicts  $\|\tilde{H}\|_{\text{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \rightarrow 0$ , consistent with the observed decrease.

*Remark 77* (Row leverage). The maximum *normalized row leverage*  $\rho(n) = \sum_{p:n \in \Omega_p} 1/D_p$  is bounded near 1.5 at all tested sizes, dominated by one singleton column ( $1/D_p \approx 1$ ) and one or two doublet columns ( $1/D_p \approx 0.5$ ). No singleton collisions (two singletons at the same  $m$ ) occur up to  $N = 4096$ . By the Gram identity  $H + I = U^T U$  where  $U$  is the  $N \times K$  normalized incidence matrix,  $\|H\|_{\text{op}} + 1 \leq \max_n \rho(n) \cdot K$  is a trivial upper bound. A tighter bound via  $\max_n \rho(n)$  governs the localized spectral spikes; if  $\rho(n) \leq C$  uniformly, then  $\|H\|_{\text{op}} \leq C - 1$ .

## 12.12 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) = \frac{1}{s_i} \sum_{p \in \mathcal{B}_i} (\log p) \left( X_p(m) - \frac{A_p}{N} \right), \quad 1 \leq i \leq L, \quad (25)$$

where each  $Y_i$  is a centered function on  $I_N$  with  $\sum_m Y_i(m) = 0$ . Let  $Y$  be the  $N \times L$  matrix with columns  $Y_i$ . Then

$$\boxed{I_L + \tilde{H} = Y^T Y \succeq 0}, \quad (26)$$

or entrywise,  $\tilde{H}_{i,j} = \langle Y_i, Y_j \rangle - \delta_{ij}$ . This is the key structural identity:  $I + \tilde{H}$  is the Gram matrix of  $L$  vectors in  $\mathbb{R}^N$ , so every eigenvalue of  $\tilde{H}$  is at least  $-1$ . Only the positive edge requires an arithmetic bound.

The Rayleigh identity (24) now reads

$$R_v = a^T \tilde{H} a = \left\| \sum_i a_i Y_i \right\|_2^2 - 1.$$

Thus  $R_v = O(1)$  is equivalent to approximate orthonormality of the  $L$  aggregate functions  $Y_i$ .

*Remark 78* (Low-to-high transfer). Let  $P = EE^T$  denote orthogonal projection onto the  $L$ -dimensional block-smooth subspace. Then  $PHu = E\tilde{H}a$  and

$$\|Hu\|^2 = \|\tilde{H}a\|^2 + \|(I - P)HEa\|^2. \quad (27)$$

At  $N = 4096$ , the data give  $\|Hu\| \approx 0.31$  and  $\|PHu\| \leq \|\tilde{H}\|_{\text{op}} \approx 0.04$ , so  $\|(I - P)Hu\| \geq 0.307$ . Almost all of  $Hu$  lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

### 12.13 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.8) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity  $\mathbf{E}[\phi_p \phi_p^T] = \frac{1}{N-1} P_0$ , where  $P_0 = I - \frac{1}{N} \mathbf{1}\mathbf{1}^T$ . For distinct primes,  $\mathbf{E}[H_{p,q}^2] = \frac{1}{N-1}$ .

**Proposition 79** (Compressed second moments). *In the independent fixed-margin model,  $\mathbf{E}[\tilde{H}_{i,j}] = 0$  for all  $i, j$ , and*

$$\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \quad (i \neq j), \quad (28)$$

$$\mathbf{E}[\tilde{H}_{i,i}^2] = \frac{2}{N-1} (1 - \kappa_i), \quad \kappa_i = \sum_{p \in \mathcal{B}_i} e_i(p)^4. \quad (29)$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E}\|\tilde{H}\|_F^2 = \frac{L^2 + L - 2 \sum_i \kappa_i}{N-1} \asymp \frac{L^2}{N}. \quad (30)$$

*Proof.* For  $i \neq j$ , disjoint block supports give  $\tilde{H}_{i,j} = \sum_{p \in \mathcal{B}_i} \sum_{q \in \mathcal{B}_j} e_i(p) e_j(q) H_{p,q}$ . Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive:  $\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1} \sum_p e_i(p)^2 \sum_q e_j(q)^2 = \frac{1}{N-1}$ . The diagonal entries involve  $\sum_{p < q \in \mathcal{B}_i} e_i(p)^2 e_i(q)^2$ , giving the  $\kappa_i$  correction.  $\square$

This proves  $\|\tilde{H}\|_{\text{op}} \rightarrow 0$  in the random model whenever  $L = o(\sqrt{N})$ . The GOE refinement predicts  $\|\tilde{H}\|_{\text{op}} \approx 2\sqrt{L/N}$ , with the factor 2 being the asymptotic edge constant.

*Remark 80* (Random surrogate comparison). For each  $N$ , we generate 20 random palindromic zero sets of the same sizes  $|Z_p|$ , build the corresponding  $\tilde{H}$ , and compare  $\|\tilde{H}\|_{\text{op}}$ . At  $N = 2048$ : actual = 0.097, random mean = 0.097,  $z$ -score =  $-0.01$ . At  $N = 1024$ : actual = 0.111, random mean = 0.130,  $z$ -score =  $-0.91$ . The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

| $N$  | $L$ | $\sqrt{L/N}$ | $\ \tilde{H}\ _{\text{op}}$ | $\ \tilde{H}\ _{\text{op}}/\sqrt{L/N}$ |
|------|-----|--------------|-----------------------------|----------------------------------------|
| 512  | 6   | 0.108        | 0.113                       | 1.04                                   |
| 1024 | 6   | 0.077        | 0.111                       | 1.45                                   |
| 2048 | 7   | 0.058        | 0.097                       | 1.66                                   |
| 4096 | 7   | 0.041        | 0.040                       | 0.97                                   |
| 8192 | 8   | 0.031        | 0.052                       | 1.67                                   |

Table 4: The compressed operator norm tracks the random CRT prediction  $\sqrt{L/N}$  with ratio  $\approx 1$ –1.7.

### 12.14 AMTD: the precise remaining conjecture

The dispersion target (Hypothesis 66) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function  $1_{\mathcal{Z}_p}$  varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:

**Hypothesis 81** (Apéry Moving-Target Dispersion (AMTD)). *There exists an absolute constant  $C$  such that, for every dyadic interval  $I_N = (N, 2N]$  and every dyadic prime block  $P < p \leq 2P$  with  $P > \sqrt{N}$ ,*

$$V^\circ(P, N) \leq C N^{o(1)} S(P, N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{\substack{P < p \leq 2P \\ 1 \leq a < p \\ x < a/p \leq x+1/N}} \frac{(\log p) F_p(a)}{p} \right|^2 dx \ll \frac{N^{o(1)}}{N} \sum_{P < p \leq 2P} \sum_{a=1}^{p-1} \left| \frac{(\log p) F_p(a)}{p} \right|^2, \quad (31)$$

where  $F_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$ . By Gallagher’s lemma, (31) implies AMTD block by block.

The classical additive large sieve gives the right-hand side of (31) multiplied by  $(1 + P^2/N)$ . For  $P > \sqrt{N}$ , this factor exceeds 2 and grows to  $O(N)$  at the top block  $P \sim N$ . The  $P^2$  barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (31) must therefore exploit the special structure of the Apéry Fourier coefficients  $F_p(a)$ —their palindromic phase constraint  $(e_p(a/2) F_p(a) \in \mathbb{R})$ , their connection to the  $\text{Sym}^2$  K3 Hasse polynomial, or the arithmetic of the zero sets  $\mathcal{Z}_p$  themselves.

*Remark 82* (Partial large-sieve result for small primes). For dyadic blocks with  $P \leq \sqrt{N}$ , the classical additive large sieve gives  $V^\circ(P, N) \leq (N + 4P^2) \sum_{P < p \leq 2P} w_p^2 Z(p)/p \leq 5 S(P, N)$ . Thus AMTD holds *unconditionally* for primes below  $\sqrt{N}$  with constant  $C = 5$ . The open case is  $P > \sqrt{N}$ , where the large-sieve factor  $1 + 4P^2/N$  exceeds any fixed constant. At the top block  $P \sim N$ , this factor is  $O(N)$ , and the generic large sieve provides no useful bound.

*Remark 83* (Linnik dispersion route). The most plausible analytic pathway is Linnik’s dispersion method: expand the  $L^2$ -discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates à la Kowalski–Michel–Sawin [10] and Drappeau [8]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient  $a \mapsto \hat{Z}_p(a) = \sum_{r \in \mathcal{Z}_p} e_p(ar)$  or the phase  $m \mapsto e_p(u b_m)$  must belong to a uniformly controlled trace-function family. The obstacle is that  $b_m \bmod p$  is computed from a nonautonomous recurrence whose complexity grows with  $p$ , and no fixed bounded-conductor  $\ell$ -adic sheaf

with trace function  $A_p(t)$  uniformly in  $p$  is currently known. The closest available framework is Forey–Fresán–Kowalski [11], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs.  $\ell$ -adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (31) already suffices for the competition target.

*Remark 84* (Palindromic Fourier factoring). For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p \in \mathcal{D}} [e(\theta m_1(p)) + e(\theta m_2(p))] = 2 \sum_{p \in \mathcal{D}} e\left(\theta \frac{3p-1}{2}\right) \cos(\pi \theta h_p),$$

where  $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$  is the doublet gap. By Parseval,  $\int_0^1 |S(\theta)|^2 d\theta = T + F$ . The diagonal AMTD requires this integral to be  $O(T)$ , equivalently that  $S(\theta)$  has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase  $e(3p\theta/2)$  modulated by the cosine  $\cos(\pi\theta h_p)$ . The gaps  $h_p$  are determined by the first zero  $r_p$  of  $b_n \bmod p$ . A proof of diagonal AMTD via this route would establish cancellation in  $S(\theta)$  by exploiting the “pseudorandom” distribution of  $h_p$  as  $p$  varies.

*Remark 85* (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

- (A) **Full operator bound:**  $\|H\|_{\text{op}} = O(1)$ . This implies a Bessel inequality for all prime columns and is far stronger than needed.
- (B) **Compressed Bessel bound:**  $\|\tilde{H}\|_{\text{op}} = O(1)$ . This asks for approximate orthonormality of only  $L = O(\log N)$  block-aggregate functions, and suffices for the all-index GCD estimate.
- (C) **Scalar Rayleigh bound:**  $a^T \tilde{H} a = O(1)$ . This is the weakest statement that closes the all-index theorem, equivalent to  $V^\circ \ll S$ . It preserves all useful signs and remains a two-prime problem.

Each implication (A) $\Rightarrow$ (B) $\Rightarrow$ (C) is strict. The random CRT model predicts  $\|\tilde{H}\|_{\text{op}} \asymp \sqrt{L/N} \rightarrow 0$  and  $|a^T \tilde{H} a| \asymp N^{-1/2}$ , both of which are far stronger than (B) and (C).

*Remark 86* (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the uncentered block load  $F_i(m) = \sum_{p \in \mathcal{B}_i} (\log p) g_p(m) = s_i Y_i(m)$ , so that  $V^\circ = \|\sum_i F_i\|_2^2$ . If each dyadic block satisfies

$$\|F_i\|_2^2 \leq C s_i^2 \quad (\text{Diagonal AMTD}), \tag{32}$$

equivalently  $1 + \tilde{H}_{ii} \leq C$ , then by Cauchy–Schwarz,

$$V^\circ \leq \left( \sum_i \|F_i\|_2 \right)^2 \leq L \sum_i \|F_i\|_2^2 \leq C L S.$$

Since  $L = O(\log N) = N^{o(1)}$ , this gives  $V^\circ \leq N^{o(1)} S$ , which suffices for  $\max_m |B(m) - \mu| \leq \sqrt{V^\circ} = O(\sqrt{N} (\log N)^{3/2}) = o(N/\log N)$ . The within-block estimate (32) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block  $P \sim N$ , it reduces to a sparse translated-set collision problem: bounding the additive energy of  $\{p + r : r \in \mathcal{Z}_p\}$  for primes  $p$  in a dyadic range.

**Block-level data at  $N = 4096$ .** The ratio  $\|F_i\|_2^2/s_i^2$  for each dyadic block at  $N = 4096$ :

| Block    | $K_i$ | $\ F_i\ _2^2/s_i^2$ | $\tilde{H}_{ii}$ |
|----------|-------|---------------------|------------------|
| $2^6$    | 2     | 1.018               | +0.018           |
| $2^7$    | 13    | 0.998               | −0.002           |
| $2^8$    | 17    | 0.987               | −0.013           |
| $2^9$    | 30    | 0.988               | −0.012           |
| $2^{10}$ | 47    | 1.015               | +0.015           |
| $2^{11}$ | 94    | 1.016               | +0.016           |
| $2^{12}$ | 177   | 0.989               | −0.011           |

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is  $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$ , confirming  $V^\circ/S = 1.002$  while the Cauchy bound gives  $V^\circ \leq 7S$ . At  $N = 8192$  the maximum block ratio remains 1.029 (block  $2^{10}$ ) with  $L = 8$  blocks. At  $N = 32,768$  the collision count for top-block doublets is  $F = 38$  with  $T = 925$ , giving  $F/T = 0.041$  (decreasing from 0.055 at  $N = 8192$ ), confirming the conjecture across all tested sizes.

*Remark 87* (Translation variance). For each pair of distinct primes  $p, q$ , averaging over all translates  $x \bmod pq$  gives

$$\frac{1}{pq} \sum_{x \bmod pq} |T_{p,q}(x; N)|^2 \leq \frac{N Z(p)^2 Z(q)^2}{pq},$$

where  $T_{p,q}(x; L) = \sum_{j=1}^L f_p(x+j) f_q(x+j)$ . For bounded zero counts  $Z(p) = O(1)$ , this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate  $x = N$  simultaneously for all prime pairs. Summing individual pair bounds absolutely costs  $O(N^{3/2})$ , which exceeds  $S \asymp N(\log N)^2$ . The  $P^2$  large-sieve barrier arises because the Fourier support of  $f_p$  has  $\sim p$  nonzero frequencies (not  $Z(p)$ ), and palindromic symmetry improves only absolute constants.

### 12.15 Poisson variance formulation

The diagonal AMTD (32) admits a clean probabilistic reformulation. For a dyadic block  $\mathcal{B}_i$ , define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p \in \mathcal{B}_i} X_p(m), \quad m \in I_N.$$

Its empirical mean and variance over  $I_N$  are

$$\mu_i = \frac{1}{N} \sum_m \lambda_i(m) = \frac{1}{N} \sum_{p \in \mathcal{B}_i} A_p, \quad V_i = \frac{1}{N} \sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N V_i = \sum_{p \in \mathcal{B}_i} D_p + \sum_{\substack{p, q \in \mathcal{B}_i \\ p \neq q}} C_{p,q}.$$

Since all primes in a dyadic block satisfy  $\log p \asymp \log P$ , one has  $\|F_i\|_2^2 \asymp (\log P)^2 N V_i$  and  $s_i^2 \asymp (\log P)^2 \sum_p D_p$ , so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight  $\log p$ ) to:

$$V_i \leq C \mu_i \quad (\text{sub-Poisson variance}). \quad (33)$$

If the indicators  $X_p(m)$  were independent Bernoulli variables (with  $\Pr[X_p = 1] = A_p/N$ ), each  $\lambda_i(m)$  would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be  $\leq 1$ . Condition (33) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

| $N$  | $K_{\text{top}}$ | $\mu$ | $V/\mu$ | collisions | Poisson pred. |
|------|------------------|-------|---------|------------|---------------|
| 1024 | 47               | 0.086 | 1.073   | 14         | 7.6           |
| 2048 | 94               | 0.079 | 0.983   | 10         | 12.8          |
| 4096 | 177              | 0.079 | 0.989   | 22         | 25.6          |
| 8192 | 363              | 0.080 | 1.011   | 60         | 52.9          |

Table 5: Column multiplicity statistics for the top dyadic block.  $\mu = \frac{1}{N} \sum_p A_p$  is the mean,  $V/\mu$  is the variance-to-mean ratio (Poisson prediction: 1), “collisions” is  $\sum_m \lambda(\lambda - 1)$ , and “Poisson pred.” is  $T^2/N$  where  $T = \sum_p A_p$ .

*Remark 88* (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p \neq q \in \mathcal{B}_i} C_{p,q} = \sum_m \lambda_i(m)(\lambda_i(m) - 1) - \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ( $P \sim N$ ), each prime has  $A_p \leq Z(p) \leq 3$  hits, so  $\lambda_i(m) \leq 3$  for all  $m$  at  $N = 8192$ . The collision count fluctuates around the Poisson prediction  $T^2/N$ , confirming the near-independence of the indicators.

*Remark 89* (Doublet dominance). Palindromic symmetry  $b_{p-1-j} \equiv b_j \pmod{p}$  forces the zero set  $\mathcal{Z}_p$  to be closed under  $r \mapsto p - 1 - r$ . If  $Z(p) > 0$ , then  $Z(p)$  is typically even, and indeed at the top block,  $Z(p) \geq 2$  for all active primes. The distribution at  $N = 8192$  is: *doublets*  $Z = 2$  (78%), *quadruplets*  $Z = 4$  (19%), *sextuplets*  $Z = 6$  (3%), *octuplets*  $Z = 8$  ( $< 1\%$ ). No singleton primes appear ( $Z = 1$  requires  $b_{(p-1)/2} \equiv 0 \pmod{p}$ , which is extremely rare for large  $p$ ).

For a doublet prime with  $\mathcal{Z}_p = \{r_p, p - 1 - r_p\}$ , the two hit positions are

$$m_1(p) = p + r_p, \quad m_2(p) = 2p - 1 - r_p,$$

satisfying  $m_1(p) + m_2(p) = 3p - 1$ . A collision between two doublet primes  $p, q$  requires one of four types:

$$\begin{aligned} \text{(A)} \quad m_1(p) = m_1(q) &\iff r_q = r_p + (p - q), \\ \text{(B)} \quad m_1(p) = m_2(q) &\iff r_p + r_q = 2q - 1 - p, \\ \text{(C)} \quad m_2(p) = m_1(q) &\iff r_p + r_q = 2p - 1 - q, \\ \text{(D)} \quad m_2(p) = m_2(q) &\iff r_q = r_p + 2(q - p). \end{aligned}$$

For  $q > p$ , type (B) is impossible:  $m_1(p) = p + r_p < 3p/2 < 3q/2 < 2q - 1 - r_q = m_2(q)$ . The collision energy therefore decomposes into three, not four, pieces:

$$F = F_A + F_C + F_D, \tag{34}$$

where  $F_A = \sum_m \nu_1(m)(\nu_1(m)-1)$ ,  $F_D = \sum_m \nu_2(m)(\nu_2(m)-1)$ ,  $F_C = 2 \sum_m \nu_1(m)\nu_2(m)$ , with  $\nu_1(m) = \#\{p : m_1(p) = m\}$  and  $\nu_2(m) = \#\{p : m_2(p) = m\}$ .

Two distinct doublet columns  $\{m_1(p), m_2(p)\}$  and  $\{m_1(q), m_2(q)\}$  can share at most one element: if both elements coincided, then  $m_1(p) + m_2(p) = m_1(q) + m_2(q)$ , i.e.,  $3p - 1 = 3q - 1$ , forcing  $p = q$ . Thus  $J_{p,q} := |\Omega_p \cap \Omega_q| \leq 1$  for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count  $F = \sum_m \lambda(m)(\lambda(m) - 1)$ . The exact variance identity is

$$V = T + F - T^2/N.$$

Since  $T \asymp N/\log N$  and  $T^2/N = o(T)$ , the diagonal AMTD ( $V \leq CT$ ) is equivalent to the collision bound  $F = O(T)$ .

### Collision decomposition.

| $N$   | $K$  | $T$ | $T^2/N$ | $F$ | $F_A$ | $F_C$ | $F_D$ | $F/T$ |
|-------|------|-----|---------|-----|-------|-------|-------|-------|
| 1024  | 47   | 39  | 1.5     | 2   | 2     | 0     | 0     | 0.051 |
| 2048  | 93   | 70  | 2.4     | 4   | 2     | 2     | 0     | 0.057 |
| 4096  | 177  | 140 | 4.8     | 0   | 0     | 0     | 0     | 0.000 |
| 8192  | 363  | 293 | 10.5    | 16  | 10    | 4     | 2     | 0.055 |
| 16384 | 669  | 528 | 17.0    | 22  | 16    | 6     | 0     | 0.042 |
| 32768 | 1145 | 925 | 26.1    | 38  | 18    | 20    | 0     | 0.041 |

The ratio  $F/T$  is consistently small and decreasing, empirically confirming  $F = o(T)$ . Under the FPI prediction  $F \approx 3K^2/N$ , one expects  $F/T \approx 3/(6.6 \log N)$ , giving 0.044 at  $N = 32,768$ , in good agreement with the observed 0.041. At all sizes  $\max_m \lambda(m) \leq 3$ .

*Remark 90* (Two-prime divisor formulation). At the top block ( $P > N$ ), each hit  $m \in \Omega_p$  implies  $p \mid b_m$  by the Lucas congruence. Therefore

$$\lambda_B(m) = \omega_{(N, 2N]}(b_m) := \#\{p \text{ prime} : N < p \leq 2N, p \mid b_m\},$$

and the collision count becomes

$$F = \sum_{m \in I_N} \omega_{(N, 2N]}(b_m)(\omega_{(N, 2N]}(b_m) - 1).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over  $m$ , the Apéry number  $b_m$  has at most  $O(1)$  prime divisors in the dyadic range  $(N, 2N]$ .

This formulation reveals the arithmetic core: the collision bound  $F = O(T)$  is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently,  $T = O(N/\log N)$ ) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events  $p \mid b_m$  and  $q \mid b_m$  are approximately uncorrelated as  $p, q$  range over a dyadic block.

**Vertical vs. horizontal projections.** The incidence graph has primes on one side and integers on the other; an edge  $(p, n)$  records  $p \mid b_n$ . Grouping by the integer index  $n$  gives the *vertical* prime-divisor count  $W_N(n) = \omega_{(N, 2N]}(b_n)$ ,  $n \in [1, N]$ . Grouping by the hit position  $m = n + p$  gives the *horizontal* column multiplicity  $\lambda(m)$ . These are different projections. By the Lucas congruence  $b_{p+r} \equiv 5b_r \pmod{p}$  (for  $r < p$ ), the horizontal multiplicity  $\lambda(m) = \omega_{(N, m]}(b_m)$ —it counts large prime divisors of  $b_m$  itself. Crucially, even the pointwise bound  $W_N(n) \leq 1$  for *every*  $n \leq N$  does not control  $\lambda(m)$ : many distinct  $(n_i, p_i)$  pairs can satisfy  $n_i + p_i = m$  with  $p_i \mid b_{n_i}$ , all contributing *different* prime divisors of the single integer  $b_m$ . The collision bound  $F = O(T)$  is therefore a genuinely horizontal, two-prime-divisor assertion about  $b_m$ .

*Remark 91* (Why classical sieves do not close the argument). The collision bound  $F = O(T)$  has the form of a two-dimensional upper-bound sieve (Selberg, Rosser–Iwaniec, Brun, GPY), but no classical sieve theorem deduces it from the one-prime data alone. After weight expansion, every such sieve introduces the counts  $A_{pq}(N) = \#\{m : pq \mid b_m\}$  for products  $pq \asymp N^2$ . Since the observation interval has length only  $N$ , each individual expected count is  $O(1/N)$  and only an *averaged* large-modulus discrepancy theorem can hold. The difficulty is intrinsic: even the strongest conceivable one-prime bound  $A_p(N) \leq 1$  for every  $p$  does not control cross-prime alignment. An adversarial configuration with all zero sets concentrating at a single index  $m_0$  satisfies  $A_p = 1$  while  $F \asymp (N/\log N)^2$ . Thus the sieve machinery can *package* a two-prime distribution theorem as a collision bound, but it cannot *manufacture* that theorem from one-prime information. The minimum new input is the bilinear estimate

$$\sum_{\substack{p \neq q \\ p, q \in (N, 2N]}} A_{pq}(N) \ll T + \frac{T^2}{N}, \quad (35)$$

which is precisely the block second-factorial-moment conjecture (diagonal AMTD, Hypothesis 81).

**Proposition 92** (Conditional collision bound). *Assume the following Frobenius Pair Independence hypothesis (FPI): for distinct doublet primes  $p, q \in (N, 2N]$ , the normalized first zeros  $r_p/p$  and  $r_q/q$  are asymptotically independent and each approximately uniform on  $[0, \frac{1}{2}]$  in the sense that, for any integers  $a \in [1, (p-1)/2]$  and  $b \in [1, (q-1)/2]$ ,*

$$\frac{1}{K(K-1)} \#\{(p, q) \text{ distinct doublet primes in } (N, 2N] : r_p = a, r_q = b\} = (1 + o(1)) \frac{4}{pq}, \quad (36)$$

where  $K$  is the number of doublet primes. Then  $F = o(T)$  and the all-index GCD theorem  $G_n = e^{o(n)}$  holds for all  $n$ .

*Proof.* We bound each piece of the decomposition (34) separately.

**Type A.** The type-A collision count is

$$F_A = \#\{(p, q) \text{ distinct doublet primes} : r_q = r_p + (p - q)\}.$$

For a fixed ordered pair  $(p, q)$ , the collision requires  $r_q = r_p + (p - q)$ . Summing the FPI probability over all valid values of  $r_p$ , the conditional probability is

$$\Pr[\text{type A} \mid p, q] = \sum_{r=1}^{(p-1)/2} \frac{2}{p} \cdot \frac{2}{q} \cdot \mathbf{1}[r + (p-q) \in [1, (q-1)/2]] \leq \frac{4 \min((p-1)/2, (3q-1)/2 - p)}{pq}.$$

For primes  $p, q \in (N, 2N]$ , this is at most  $4/N$  when  $|p - q| \leq N/2$ , and vanishes when  $|p - q| > N$ . Summing over all  $K(K-1)$  ordered pairs:

$$F_A \leq K(K-1) \cdot \frac{4}{N} \cdot (1 + o(1)) \leq \frac{4K^2}{N} \cdot (1 + o(1)).$$

Since  $K \asymp N/\log N$  and  $T = 2K + O(K^{1/2})$ , we obtain  $F_A \leq 4T^2/(4N) \cdot (1 + o(1)) = T^2/N \cdot (1 + o(1)) = o(T)$ .

**Type C.** The collision condition  $r_p + r_q = 2p - 1 - q$  constrains a sum of independent uniforms. Under FPI,  $r_p + r_q$  has a trapezoidal distribution on  $[2, (p + q - 2)/2]$  with maximum density

$2/\min(p, q) \leq 2/N$ . The target  $2p - 1 - q$  lies in the support when  $q < 2p - 1$ , which holds for  $p > (q + 1)/2$ . Thus  $\Pr[\text{type C} \mid p, q] \leq 2/N$ , and  $F_C \leq 2K^2/N \cdot (1 + o(1)) = o(T)$ .

**Type D.** The condition  $r_q = r_p + 2(q - p)$  has the same structure as type A with gap  $2(q - p)$  in place of  $p - q$ . By the same argument,  $F_D \leq 4K^2/N \cdot (1 + o(1)) = o(T)$ .

Combining all three,  $F = F_A + F_C + F_D = o(T)$ . The diagonal AMTD  $\|F_i\|_2^2 \leq C s_i^2$  follows with  $C = 1 + o(1)$  at each dyadic block, giving  $V^\circ \leq (1 + o(1)) L S = O(\log N) S = N^{o(1)} S$ .  $\square$

*Remark 93* (Status of the FPI hypothesis). The hypothesis (36) is a pair decorrelation statement for the zero positions across different primes. The marginal equidistribution of  $r_p/p$  is supported by the Kummer structure of the reduced generating series (Caruso–Fürnsinn–Vargas–Montoya–Zudilin [7]), which shows that the Galois group of  $f_\alpha \bmod p$  is cyclic of order  $(p - 1)/\gcd(2, p - 1)$ . The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of  $\text{Sym}^2 \text{SL}_2 \simeq \text{SO}_3$  type.

However, neither the cyclic Kummer group nor the  $\text{SO}_3$  monodromy controls the zero *positions*  $r_p$ . To see why, observe that the zero-set  $\mathcal{Z}_p = \{j < p : p \mid b_j\}$  is a *spectral* zero-set: writing  $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$  and  $f_p(u) = A_p(g^u)$  for a primitive root  $g$ , multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u) g^{-ju} = -b_j,$$

so  $j \in \mathcal{Z}_p$  iff the  $j$ -th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the *roots* of  $A_p(t)$  in  $\mathbb{F}_p$  are evaluation zeros, controlled by the Kummer factorization  $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$ . The Mellin coefficient index  $j$  and the discrete logarithm  $\text{ind}_g(x)$  of a root  $x$  live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (36) is therefore a genuinely new hypothesis: it concerns the spectral zero process  $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j \equiv 0\}}$ , not a Frobenius class function or a discrete logarithm. Neither Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N < p, q \leq 2N \\ p \neq q}} \sum_{m \in I_N} X_p(m) X_q(m) \ll T_N + \frac{T_N^2}{N}, \quad (37)$$

where  $X_p(m) = \mathbf{1}_{\{p \leq m, b_{m-p} \equiv 0 \pmod{p}\}}$ . The left side equals the ordered collision count  $F_N^{\text{ord}}$ . Via the exponential-sum detection  $\mathcal{A}_p(j) = p^{-1} \sum_{a \bmod p} e_p(a b_j)$ , the conjecture (37) reduces to cancellation in the mixed-modulus sum

$$\sum_{p \neq q} \frac{1}{pq} \sum_{\substack{a \bmod p \\ c \bmod q}}' \sum_{m \in I_N} e_p(a b_{m-p}) e_q(c b_{m-q}), \quad (38)$$

where  $\sum'$  omits the term  $a = c = 0$  (which produces the main term). The inner sum is over a nonautonomous recurrence orbit:  $m \mapsto (b_{m-p}, b_{m-q})$ . Standard Weil bounds do not apply.

A proof of (37) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks  $b_j \bmod p$ ; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at  $1/p$  scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ( $F/T \leq 0.041$  at  $N = 32,768$ ).

**Proposition 94** (Profile decomposition). *For each prime  $p \in (N, 2N]$ , let  $A_p = |\Omega_p \cap I_N|$  and  $L_p = 2N - p$ . Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N < p \leq 2N \\ p < m}} \frac{A_p}{L_p}, \quad m \in I_N.$$

*Then:*

- (i)  $\sum_{m \in I_N} \rho(m) = T_N$ ;
- (ii)  $\|\rho\|_2^2 := \sum_{m \in I_N} \rho(m)^2 \leq T_N^2/N$ ;
- (iii)  $F_N \leq 2\|\rho\|_2^2 + 2\|\lambda - \rho\|_2^2 - T_N \leq 2T_N^2/N + 2\|\lambda - \rho\|_2^2$ .

*In particular, the residual dispersion estimate  $\|\lambda - \rho\|_2^2 \ll T_N$  would imply  $F_N = O(T_N)$ .*

*Proof.* Part (i):  $\sum_m \rho(m) = \sum_p (A_p/L_p) \cdot L_p = \sum_p A_p = T_N$ . For (ii), write  $\|\rho\|_2^2 = \sum_{p,q} (A_p A_q)/(L_p L_q) \cdot (2N - \max(p, q))$ . By the PNT approximation  $\rho(m) \approx (2/\log N) \log(N/(2N-m))$  and the integral  $\int_0^1 (\log 1/t)^2 dt = 2$ , one obtains  $\|\rho\|_2^2 \sim c T_N^2/N$  for an effective constant  $c$ . Numerically,  $\|\rho\|_2^2/(T_N^2/N)$  stabilizes near 1.25 for  $N = 512, \dots, 32,768$  (Table 6). Part (iii) follows from  $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$  applied to  $\lambda = \rho + \eta$ .  $\square$

**Remark 95** (Orthogonal decomposition). Setting  $\eta = \lambda - \rho$ , the identity  $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + \|\eta\|_2^2$  gives

$$F = \|\rho\|_2^2 + 2\langle \rho, \eta \rangle + (\|\eta\|_2^2 - T). \quad (39)$$

Under the independent profile model (each  $X_p(m)$  is an independent Bernoulli with mean  $\pi_p(m) = A_p/L_p$  when  $p < m$ ), the three terms in (39) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2/N)$ : the deterministic profile energy, tending to 0 relative to  $T$ .
- $\|\eta\|_2^2 \approx T$ : the shot-noise variance of  $T$  independent Bernoulli hits.
- $\langle \rho, \eta \rangle \approx 0$ : the profile and residual are decorrelated.

Hence  $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$ . Computation (Table 6) confirms this decomposition:  $\|\eta\|_2^2/T$  is stable near 1.0 (matching palindromic random surrogates), while  $F \approx \|\rho\|_2^2$  to within 1% at  $N = 16,384$  and within 8% at  $N = 32,768$ . The cross-term  $\langle \rho, \eta \rangle$  is  $O(1)$  at every tested scale (e.g.  $-0.03$  at  $N = 16,384$  and  $0.40$  at  $N = 32,768$ , versus  $T = 664$  and  $1182$ ). Moreover, the cross-term has a *deterministic* bound:  $|\langle \rho, \eta \rangle| \leq \rho_{\max} \cdot T$  where  $\rho_{\max} = \max_m \rho(m) = \sum_p A_p/(2N-p)$ . Since  $\rho_{\max}/(T/N)$  stabilizes near 1.8 (Table 6),  $|\langle \rho, \eta \rangle| \leq 1.8 T^2/N = o(T)$ . The unconditional collision theorem  $F = O(T)$  is therefore equivalent to the shot-noise estimate  $\|\eta\|_2^2 = O(T)$ , which is the residualized AMTD (Hypothesis 81).

**Remark 96** (Exact profile constant). The column  $\|\rho\|_2^2/(T^2/N)$  in Table 6 converges to a constant  $c_2$  as  $N \rightarrow \infty$ , with exact closed form

$$c_2 = \frac{2(\ln 2)^2 - 4 \ln 2 + 2}{(2 \ln 2 - 1)^2} = 1.261982 \dots \quad (40)$$

Table 6: Profile decomposition diagnostics for the full hit measure  $\mu_N$ . All norms over  $I_N = (N, 2N]$ . “Surr” denotes the mean of  $\|\eta\|_2^2/T$  over 10 palindromic random surrogates.

| $N$   | $T$  | $F$ | $\ \rho\ _2^2$ | $\ \rho\ _2^2/(T^2/N)$ | $\ \eta\ _2^2/T$ | Surr |
|-------|------|-----|----------------|------------------------|------------------|------|
| 512   | 21   | 2   | 1.1            | 1.28                   | 1.05             | 0.98 |
| 1024  | 50   | 2   | 3.0            | 1.22                   | 0.98             | 0.99 |
| 2048  | 86   | 4   | 4.5            | 1.26                   | 0.99             | 1.01 |
| 4096  | 172  | 2   | 9.4            | 1.31                   | 0.96             | 1.00 |
| 8192  | 354  | 20  | 19.6           | 1.28                   | 1.00             | 1.00 |
| 16384 | 664  | 34  | 33.7           | 1.25                   | 1.00             | 1.01 |
| 32768 | 1182 | 58  | 53.8           | 1.26                   | 1.00             | —    |

The derivation is direct. The triangular profile  $\rho(m) = \sum_{N < p \leq m} Z(p)/p$  for  $m \in I_N$  has the continuous approximation  $\rho(Nt) \approx (\ln t)/(\ln N)$  for  $t \in [1, 2]$ , by Mertens’ theorem. Then

$$T = \sum_m \rho(m) \approx \frac{N}{\ln N} \int_1^2 \ln t \, dt = \frac{N}{\ln N} (2 \ln 2 - 1),$$

$$\|\rho\|_2^2 = \sum_m \rho(m)^2 \approx \frac{N}{(\ln N)^2} \int_1^2 (\ln t)^2 \, dt = \frac{N}{(\ln N)^2} (2(\ln 2)^2 - 4 \ln 2 + 2).$$

The ratio gives (40). The exact palindromic null expectation confirms the match: at  $N = 1024$  the finite- $N$  ratio is 1.209, at  $N = 4096$  it is 1.281, at  $N = 16,384$  it is 1.254, oscillating around  $c_2$  with  $O(1/\ln N)$  corrections from the prime number theorem. At all tested sizes,  $E_{\text{null}}[F]$  agrees with the observed  $F$  to within the expected stochastic fluctuation.

*Remark 97* (Higher factorial moments). The third factorial moment  $F_3 = \sum_m \lambda(m)(\lambda(m)-1)(\lambda(m)-2)$  equals 6 at both  $N = 16,384$  and  $N = 32,768$ , contributed by a single row with  $\lambda = 3$  (giving  $3 \cdot 2 \cdot 1 = 6$ ). The palindromic model predicts  $E[F_3] \approx c_3 T^3/N^2$  where  $c_3$  is a geometry-of-truncation constant. At  $N = 16,384$  with  $T = 664$ ,  $T^3/N^2 \approx 1.09$ , giving  $E[F_3] \approx 7$ —consistent with the observation. The fourth factorial moment  $F_4 = 0$  at both sizes (no row has  $\lambda \geq 4$ ), and  $\max_m \lambda(m) = 3$ .

## 12.16 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial  $N_h$  (whose roots in  $\mathbb{F}_p$  are necessary for a simultaneous zero pair  $b_x \equiv b_{x+h} \equiv 0$ ) has  $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \leq 3(h-1)$ , giving  $\sum_{h < H} r_p(h) \leq 3H^2/2$  and hence  $Z(p) \leq Cp^{2/3}$ . However, the actual root counts are drastically smaller: for all primes  $p \leq 10,000$  with  $Z(p) \geq 2$  and all gaps  $h \leq 100$ ,  $\sum_{h \leq 100} r_p(h) \leq 1$ , with most primes having  $\sum = 0$  (the zero pairs are spaced more than 100 apart).

**Hypothesis 98** (Linear root sum).  $\sum_{1 \leq h < H} r_p(h) \ll H$  for every prime  $p$  and every  $H \geq 1$ .

This would improve the zero-count to  $Z(p) \ll p^{1/2}$ , strengthening the exponent in Proposition 67 from  $N^{5/6}$  to  $N^{3/4}$ . The heuristic justification is that  $N_h$  has degree  $3(h-1)$  and (conjecturally) Galois group close to  $S_{3(h-1)}$  over  $\mathbb{Q}$ , so that by the Chebotarev density theorem  $\mathbb{E}[r_p(h)] = 1$  for each  $h$ .

## 12.17 The exact random model

The palindromic identity  $b_{p-1-j} \equiv b_j \pmod{p}$  forces the zero-set to consist of reflected pairs. Let  $V_p^+$  denote the space of inversion-invariant functions  $f: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p$ , i.e.  $f(t) = f(t^{-1})$ . The algebraic discrete Fourier transform  $\hat{f}(j) = \sum_{t \in \mathbb{F}_p^\times} f(t) t^{-j}$  maps  $V_p^+$  isomorphically onto the subspace of sequences satisfying  $\hat{f}(j) = \hat{f}(-j)$ .

**Proposition 99** (Random inversion-invariant model). *Choose  $f$  uniformly from  $V_p^+$ . Then the number  $K$  of noncentral reflected pairs  $\{j, p-1-j\}$  with  $\hat{f}(j) = 0$  satisfies*

$$K \sim \text{Binomial}\left(\frac{p-3}{2}, \frac{1}{p}\right) \xrightarrow{p \rightarrow \infty} \text{Poisson}(1/2).$$

*Proof.* The spectral coefficients of distinct reflected pairs are independent uniform elements of  $\mathbb{F}_p$  (the DFT is invertible), so each vanishes with probability  $1/p$ .  $\square$

The Apéry Hasse polynomial  $H_p(t) = \sum b_k t^k$  is a specific element of  $V_p^+$  with additional algebraic structure ( $H_p = \Delta^\varepsilon B_p^2$ ). The conjecture that  $Z(p) \sim 2 \text{Poisson}(1/2)$  is equivalent to asserting that  $H_p$  behaves, for the spectral-zero statistic, like a generic element of  $V_p^+$ .

The square structure can be incorporated directly. Let  $V_p^- = \{g: \mathbb{F}_p^\times \rightarrow \mathbb{F}_p : g(t^{-1}) = -g(t)\}$  denote the anti-invariant subspace, with  $\dim V_p^- = (p-3)/2$ . For uniform  $g \in V_p^-$ , the function  $f = g^2$  lies in  $V_p^+$  and vanishes at the two fixed points  $t = \pm 1$ .

**Proposition 100** (Random anti-invariant square model). *Choose  $g$  uniformly from  $V_p^-$  and put  $f = g^2$ . For any fixed distinct noncentral reflected pairs represented by  $j_1, \dots, j_r$ ,*

$$\Pr(\hat{f}(j_1) = \dots = \hat{f}(j_r) = 0) = p^{-r} + O_r(p^{-c_r p})$$

for some  $c_r > 0$ . In particular, the pair-zero count  $K^{\text{sq}} \Rightarrow \text{Poisson}(1/2)$ .

*Proof.* Write  $u_i = g(x_i)$  for orbit representatives. The spectral coefficient  $\hat{f}(j) = \sum_i u_i^2 (x_i^j + x_i^{-j})$  is a diagonal quadratic form  $Q_\lambda(u)$  in the independent uniform variables  $u_i$ . For  $\lambda \neq 0$ , the physical function  $h_\lambda(t) = \sum_a \lambda_a (t^{j_a} + t^{-j_a})$  has Mellin support  $\leq 2r$ , so by the finite-field Mattson–Solomon uncertainty inequality, at least  $(p-1)/(4r)-1$  orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value  $p^{-1/2}$ . The factorial-moment method then gives the result.  $\square$

Multiplication by  $\Delta(t)^\varepsilon$  affects only  $O(1)$  orbits and does not change the limiting law. Thus the  $\text{Sym}^2$  squareness of the Apéry Hasse polynomial is fully compatible with  $\text{Poisson}(1/2)$  in any random model where the square root  $B_p$  behaves generically.

**Proposition 101** (Quadratic rank and both halves of the spectrum). *Let  $B(t)$  be a uniformly random normalized reciprocal polynomial of degree  $d = (p-1-2\varepsilon)/2$  over  $\mathbb{F}_p$ , with free variables  $x_1, \dots, x_m$  where  $m = \lfloor d/2 \rfloor$ . Put  $H = \Delta^\varepsilon B^2$  and split the noncentral reflected-pair representatives into a triangular range  $\mathcal{J}_{\text{tri}}$  ( $|\mathcal{J}_{\text{tri}}| = m$ ) and a nonlinear range  $\mathcal{J}_{\text{nl}}$  ( $|\mathcal{J}_{\text{nl}}| = p/4 + O(1)$ ).*

- (i) *In the triangular range,  $H_j$  is an affine-linear function of  $(x_1, \dots, x_m)$  with a unitriangular Jacobian. Hence  $K_{\text{tri}} = \#\{j \in \mathcal{J}_{\text{tri}} : H_j = 0\} \Rightarrow \text{Poisson}(1/4)$ .*
- (ii) *In the nonlinear range, each  $H_j$  is an affine quadratic form in  $(x_1, \dots, x_m)$  whose quadratic part has rank  $\geq m/2 - O(1) \gg p$ . By the Gauss-sum bound for affine quadratic forms of rank  $R$ :  $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$ . The pencil-rank estimate from Proposition 100 gives  $K_{\text{nl}} \Rightarrow \text{Poisson}(1/4)$ .*

(iii) Without requiring independence between the two halves,

$$\mathbf{E} Z(p) = 2 \mathbf{E} K = 2(\mathbf{E} K_{\text{tri}} + \mathbf{E} K_{\text{nl}}) = 2\left(\frac{1}{4} + \frac{1}{4}\right) + o(1) = 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero  $\mathbb{F}_p$ -linear combination of  $H_{j_1}, \dots, H_{j_s}$  with at least one  $j_\nu \in \mathcal{J}_{\text{nl}}$  has quadratic rank  $\gg p/s$ —then the full pair-zero count  $K = K_{\text{tri}} + K_{\text{nl}} \Rightarrow \text{Poisson}(1/2)$ .

*Remark 102* (Separation of random model from deterministic problem). Proposition 101 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual  $B_p$  supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about  $\sum_{p \leq x} Z(p)$  requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

## 12.18 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) = \sum_{r=0}^{p-2} e_p(a(b_{r+1} - b_r)), \quad a \neq 0.$$

A nontrivial bound  $|C_p(a)| \ll p^{1-\eta}$  is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.4).

For all 666 primes  $p \leq 5000$ , we computed  $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$ . The results:

- $\max_a |C_p(a)|/\sqrt{p}$  has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$  has mean 1.23 and maximum 1.80;
- the distribution of  $\max/\sqrt{p}$  matches the Gumbel law expected for the maximum of  $p$  independent unit-variance Gaussians:  $\max \approx \sqrt{2 \log p}$ .

For shifts  $s = 2, 3, 5$ , the same behavior holds. The histogram of the differences  $b_{r+1} - b_r \pmod{p}$  has std/mean  $\approx 1$  and the fraction of zero-bins is  $\approx 1/e$ , precisely the Poisson(1) model.

**Conclusion:** the Apéry sequence  $b_r \pmod{p}$  is computationally indistinguishable from a random function  $\{0, \dots, p-1\} \rightarrow \mathbb{F}_p$  in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

## 12.19 Value-collision energy

A second pseudorandomness diagnostic is the *value-collision energy*

$$E(p) = \sum_{a \in \mathbb{F}_p} N_p(a)^2, \quad N_p(a) = \#\{j \in \{0, \dots, p-1\} : b_j \equiv a \pmod{p}\}.$$

By Cauchy–Schwarz,  $Z(p)^2 = N_p(0)^2 \leq E(p)$ , so  $E(p) \ll p$  would imply  $Z(p) \ll \sqrt{p}$ .

For a uniformly random function  $\{0, \dots, p-1\} \rightarrow \mathbb{F}_p$ , each  $N_p(a) \sim \text{Poisson}(1)$  and  $E(p)/p \rightarrow 2$ . The palindromic symmetry  $b_{p-1-j} \equiv b_j$  forces the half-sequence values to be doubled, giving

$E(p)/p \rightarrow 3$ : each half-position contributes independently to  $N_p(a)$ , and the doubling multiplies the second moment by 4 while the mean remains 1, yielding  $\mathbb{E}[N_p(a)^2] = 4(\frac{1}{2} + \frac{1}{4}) = 3$ .

We computed  $E(p)/p$  for all 1227 primes  $p \leq 10,000$ . The results confirm the palindromic prediction:  $\text{mean}(E(p)/p) = 2.996$ , with 68% of primes in  $[2.95, 3.05]$  and no dependence on  $Z(p)$ . The value-collision energy thus confirms that the Apéry sequence modulo  $p$  is anti-concentrated at the scale predicted by the palindromic random model.

## 13 Conclusion

The principal unconditional result is the quantitative exceptional-set theorem: for every fixed  $\varepsilon > 0$ ,

$$\#\{n \leq X : \log G_n > \varepsilon n\} \ll_{\varepsilon} (\log X)^2.$$

Consequently  $G_n = e^{o(n)}$  on a set of natural density one; the exceptional set has upper Banach density zero and finite harmonic weight. The arithmetic engine is the uniform interval estimate  $Z(p) \ll p^{2/3}$ , proved by the gap-polynomial and separated-block method.

The all- $n$  theorem reduces to the block second-factorial-moment estimate

$$F_N \ll T_N + \frac{T_N^2}{N}, \quad (41)$$

which asserts that prime columns in the top block do not systematically align at one integer. The profile decomposition of  $F_N$  separates this into the deterministic profile energy  $\|\rho\|^2 \sim c_2 T^2/N$  (with  $c_2$  given exactly by (40)), a deterministic cross-term  $\langle \rho, \eta \rangle = o(T)$ , and the residual shot-noise term  $\|\eta\|^2$ . The estimate (41) is equivalent to the shot-noise bound  $\|\eta\|^2 = O(T)$ , and computation through  $N = 32,768$  gives  $\|\eta\|^2/T \approx 1.0$ .

This gap is genuine. All classical sieves (Selberg, Brun, Rosser–Iwaniec, GPY) are circular for (41): they expand into the same intersection counts  $A_{pq}(N)$  that the estimate controls, and their quadratic-form optimization requires the two-prime level of distribution as input (Remark 91). On the geometric side, the K3 lisse sheaf  $R^2 f_* \mathbb{Q}_{\ell}$  has bounded rank and conductor, but the event  $p \mid b_j$  is a same-characteristic crystalline condition. The Jacobi counterexample shows that a rank-one sheaf with the same formal properties can have  $\gg p$  bad tame characters. The gap-polynomial identities are monochromatic: they control two incidences of the *same* prime and do not constrain the rainbow intersections that determine  $\|\eta\|^2$ .

Two precise sufficient criteria for Hypothesis  $Z$  emerge: a bounded-degree tame Hasse polynomial (giving  $Z(p) \leq 2D$ ), or a unique-unit-root theorem for all but  $O(1)$  tame twists. The fiberwise Hasse–Witt invariant  $A_p(t) = \sum_{n < p} b_n t^n$  is explicitly known from Stienstra–Beukers [20], but it controls which *fibers* are nonordinary, not which *Mellin coefficients* vanish; the character-twisted Hasse computation is a fundamentally different problem on the global twisted cohomology  $H_{\text{rig},c}^1(U, \mathcal{V}_{\text{tr}} \otimes \mathcal{K}_{\omega^{-j}})$ . The raw interpolation  $j \mapsto b_j$  has degree  $p - 2$  already for  $p = 11$  and  $13$ , so bounded tame Hasse degree is not automatic; it would require a nontrivial Gauss/Jacobi normalization of the first Frobenius block.

Computation through all 78,496 primes  $p \leq 10^6$  gives  $Z(p) \leq 12$  and mean  $\approx 1$ ; the value-collision energy ratio  $E(p)/p \rightarrow 3$  and the profile constant  $c_2 \rightarrow 1.262$  match the palindromic null at every tested scale. These observations support all conjectures but are not used as proofs.

The most direct path forward is an explicit Hasse–Witt calculation for the rank-three transcendental crystal, determining the slope-zero multiplicity of the Kummer-twisted compactly supported cohomology as a function of the character parameter. The Galois-group analysis of reductions of Apéry-like series by Caruso–Fürnsinn–Vargas–Montoya–Zudilin [7] may provide algebraic input,

though their results concern the fiber parameter and do not directly control the character-twisted Mellin structure or the two-prime dispersion.

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